Differential Equations Workbook Solutions

Series solutions

krista king MATH

POWER SERIES BASICS

■ 1. Where is the power series centered?

$$\sum_{n=0}^{\infty} c_n(x+3)^n = c_0 + c_1(x+3) + c_2(x+3)^2 + c_3(x+3)^3 + \ldots$$

Solution:

A power series centered at x_0 is given by

$$\sum_{n=0}^{\infty} c_n(x-x_0)^n = c_0 + c_1(x-x_0) + c_2(x-x_0)^2 + c_3(x-x_0)^3 + \ldots$$

We also call this a power series in $x - x_0$. In the given series,

$$\sum_{n=0}^{\infty} c_n(x+3)^n = c_0 + c_1(x+3) + c_2(x+3)^2 + c_3(x+3)^3 + \ldots$$

we can identify $x_0 = -3$, and say that the power series is centered at $x_0 = -3$.

■ 2. Find the radius of convergence of the power series.

$$\sum_{n=0}^{\infty} \frac{3^{2n}}{100^n}(10x)^n$$

© Krista King Math

Solution:

Let's rewrite the series.

$$\sum_{n=1}^{\infty}\left(\frac{3^2}{100}\right)^n 10^n x^n$$

$$\sum_{n=1}^{\infty}\left(\frac{3^2}{100}\cdot 10\right)^n x^n$$

$$\sum_{n=1}^{\infty}\left(\frac{9}{10}\right)^n x^n$$

Then applying the ratio test to the power series gives

$$L=\lim_{n\to\infty}\left|\frac{c_{n+1}}{c_n}\right|$$

$$L=\lim_{n\to\infty}\left|\frac{\left(\frac{9}{10}\right)^{n+1}x^{n+1}}{\left(\frac{9}{10}\right)^n x^n}\right|$$

$$L=\lim_{n\to\infty}\left|\frac{9}{10}x\right|$$

$$L=\frac{9}{10}|x|$$

© Krista King Math

To determine where the series converges absolutely, we set up the inequality $L < 1$.

$$\frac{9}{10}|x| < 1$$

$$|x| < \frac{10}{9}$$

$$-\frac{10}{9} < x < \frac{10}{9}$$

So the radius of convergence is $R = 10/9$.

■ 3. Find the Maclaurin series representation of $f(x) = \cosh(9x^2)$.

Solution:

Because we already have the Maclaurin series representation of $f(x) = \cosh x$, we can simply replace x with $9x^2$.

$$\cosh x = \sum_{n=0}^{\infty} \frac{1}{(2n)!} x^{2n}$$

$$\cosh(9x^2) = \sum_{n=0}^{\infty} \frac{1}{(2n)!} (9x^2)^{2n}$$

$$\cosh(9x^2) = \sum_{n=0}^{\infty} \frac{1}{(2n)!} (9^{2n} x^{4n})$$

© Krista King Math

$$\cosh(9x^2) = \sum_{n=0}^{\infty} \frac{9^{2n}}{(2n)!} x^{4n}$$

■ 4. Find the Maclaurin series representation of the function.

$$f(x) = 4\ln(-2 - 3x^3)$$

Solution:

Because we already have the Maclaurin series representation of $f(x) = \ln(1 + x)$, we can start by rewriting the argument so that it starts with positive 1.

$$f(x) = 4\ln(-2 + 3 - 3 - 3x^3)$$

$$f(x) = 4\ln(1 - 3 - 3x^3)$$

$$f(x) = 4\ln(1 + (-3 - 3x^3))$$

Now we'll replace x with $-3 - 3x^3$ in the Maclaurin series representation of $f(x) = \ln(1 + x)$.

$$\ln(1 + x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n$$

$$\ln(1 + (-3 - 3x^3)) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} (-3 - 3x^3)^n$$

© Krista King Math

Now we'll multiply by 4 and simplify the final Maclaurin series representation.

$$4\ln(1+(-3-3x^3)) = 4\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}(-3-3x^3)^n$$

$$4\ln(-2-3x^3) = \sum_{n=1}^{\infty}\frac{4(-1)^{n+1}}{n}((-1)(3+3x^3))^n$$

$$4\ln(-2-3x^3) = \sum_{n=1}^{\infty}\frac{4(-1)^{n+1}}{n}(-1)^n(3+3x^3)^n$$

$$4\ln(-2-3x^3) = \sum_{n=1}^{\infty}\frac{4(-1)^{2n+1}}{n}(3+3x^3)^n$$

The exponent $2n+1$ will create the a $-$, $+$, $-$, $+$, ... pattern, but the simpler exponent n accomplishes the same thing.

$$4\ln(-2-3x^3) = \sum_{n=1}^{\infty}\frac{4(-1)^{n}}{n}(3+3x^3)^n$$

■ 5. Find the Maclaurin series representation of the function.

$$f(x) = \frac{6}{\sqrt{2x}-1}$$

Solution:

© Krista King Math

Because we already have the Maclaurin series representation of $f(x) = 1/(1-x)$, we can start by rewriting the function so that the denominator starts with positive 1.

$$f(x) = -\frac{6}{1-\sqrt{2x}}$$

Now we'll replace x with $\sqrt{2x}$ in the Maclaurin series representation of $f(x) = 1/(1-x)$.

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

$$\frac{1}{1-\sqrt{2x}} = \sum_{n=0}^{\infty} (\sqrt{2x})^n$$

Now we'll multiply by -6 and simplify the final Maclaurin series representation.

$$-\frac{6}{1-\sqrt{2x}} = -6\sum_{n=0}^{\infty} (\sqrt{2x})^n$$

$$-\frac{6}{1-\sqrt{2x}} = -\sum_{n=0}^{\infty} 6((2x)^{\frac{1}{2}})^n$$

$$-\frac{6}{1-\sqrt{2x}} = -\sum_{n=0}^{\infty} 6(2x)^{\frac{n}{2}}$$

6. Find the first three derivatives of the power series.

© Krista King Math

$$f(x) = \sum_{n=0}^{\infty} 3(-1)^n (2x^2)^n$$

Solution:

To differentiate the power series representation, we can apply power rule for derivatives.

$$f(x) = \sum_{n=0}^{\infty} 3(-1)^n (2x^2)^n$$

$$f'(x) = \sum_{n=0}^{\infty} 3n(-1)^n (2x^2)^{n-1}(4x)$$

$$f'(x) = \sum_{n=0}^{\infty} 12n(-1)^n x(2x^2)^{n-1}$$

To find the second derivative, we can apply power rule again.

$$f''(x) = \sum_{n=0}^{\infty} 12n(-1)^n [(2x^2)^{n-1} + (n-1)x(2x^2)^{n-2}(4x)]$$

$$f''(x) = \sum_{n=0}^{\infty} 12n(-1)^n [(2x^2)^{n-1} + (n-1)4x^2(2x^2)^{n-2}]$$

© Krista King Math

ADDING POWER SERIES

■ 1. Are the power series in phase? Do the indices match?

$$\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} \qquad \sum_{n=0}^{\infty} \frac{(-1)^n}{9^n} x^{2n}$$

Solution:

The power series are in phase because both series begin with the x^0 term. But the indices don't match. Therefore, if we wanted to add the series, we'd need to make their indices match by substituting $k = n - 2$ and $n = k + 2$ into the first series, while substituting $k = 2n$ and $n = k/2$ into the second series.

■ 2. As written, can the power series be added?

$$\sum_{n=0}^{\infty} n!x^n \qquad 5 + \sum_{n=1}^{\infty} \frac{3^n}{n!} x^n$$

Solution:

© Krista King Math

As written, the series can't be added. They aren't in phase because the first series begins with the x^0 term, while the second series begins with the x^1 term. Furthermore, the indices of the series don't match.

To set up the series for addition, we'd pull the $n = 0$ term out of the first series to put the series in phase. Then we'd make the substitution $n = k$ into both series to make the indices match at $k = 1$.

■ 3. Find the sum.

$$\sum_{n=3}^{\infty}(2n-1)c_n x^{n-3} + \sum_{n=1}^{\infty} 2nc_n x^{n-1}$$

Solution:

Both series begin with the x^0 term, so they're already in phase. To make their indices match, we'll make the substitution $k = n - 3$ and $n = k + 3$ into the first series, and make the substitution $k = n - 1$ and $n = k + 1$ into the second series.

$$\sum_{n=3}^{\infty}(2n-1)c_n x^{n-3} + \sum_{n=1}^{\infty} 2nc_n x^{n-1}$$

$$\sum_{k=0}^{\infty}(2(k+3)-1)c_{k+3} x^{k} + \sum_{k=0}^{\infty} 2(k+1)c_{k+1} x^{k}$$

The series are now in phase with matching indices, so we'll add them and then factor out x^k.

© Krista King Math

$$\sum_{k=0}^{\infty}(2(k+3)-1)c_{k+3}x^k+2(k+1)c_{k+1}x^k$$

$$\sum_{k=0}^{\infty}[(2(k+3)-1)c_{k+3}+2(k+1)c_{k+1}]x^k$$

4. Add the power series.

$$\sum_{n=2}^{\infty}c_n n(n-1)x^{n-2} \qquad x\sum_{n=0}^{\infty}c_n x^n$$

Solution:

Rewrite the second series by bringing the x into the series representation.

$$x\sum_{n=0}^{\infty}c_n x^n$$

$$\sum_{n=0}^{\infty}c_n x^{n+1}$$

Now we can write the sum as

$$\sum_{n=2}^{\infty}c_n n(n-1)x^{n-2}+\sum_{n=0}^{\infty}c_n x^{n+1}$$

© Krista King Math

The first series begins with the x^0 term, while the second series begins with the x^1 term, so the series aren't in phase. To put them in phase, we'll pull the x^0 term out of the first series.

$$c_2 2(2-1)x^{2-2} + \sum_{n=3}^{\infty} c_n n(n-1)x^{n-2} + \sum_{n=0}^{\infty} c_n x^{n+1}$$

$$2c_2 + \sum_{n=3}^{\infty} c_n n(n-1)x^{n-2} + \sum_{n=0}^{\infty} c_n x^{n+1}$$

The series are now in phase, but their indices don't match. To make the indices match, we'll make the substitution $k = n - 2$ and $n = k + 2$ into the first series, and make the substitution $k = n + 1$ and $n = k - 1$ into the second series.

$$2c_2 + \sum_{k=1}^{\infty} c_{k+2}(k+2)(k+2-1)x^k + \sum_{k=1}^{\infty} c_{k-1}x^k$$

The series are now in phase with matching indices, so we'll add them and then factor out x^k.

$$2c_2 + \sum_{k=1}^{\infty} c_{k+2}(k+2)(k+2-1)x^k + c_{k-1}x^k$$

$$2c_2 + \sum_{k=1}^{\infty} [c_{k+2}(k+2)(k+2-1) + c_{k-1}]x^k$$

■ 5. Find the sum.

© Krista King Math

$$(x^2+1)\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + x\sum_{n=1}^{\infty} nc_n x^{n-1} - x\sum_{n=0}^{\infty} c_n x^n$$

Solution:

Rewrite the first series by splitting it into two series.

$$(x^2+1)\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + x\sum_{n=1}^{\infty} nc_n x^{n-1} - x\sum_{n=0}^{\infty} c_n x^n$$

$$x^2\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + x\sum_{n=1}^{\infty} nc_n x^{n-1} - x\sum_{n=0}^{\infty} c_n x^n$$

Now rewrite the first series by bringing the x^2 into the series representation, and rewrite the third and fourth series by bringing the x into the series representation.

$$\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2+2} + \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + \sum_{n=1}^{\infty} nc_n x^{n-1+1} - \sum_{n=0}^{\infty} c_n x^{n+1}$$

$$\sum_{n=2}^{\infty} n(n-1)c_n x^{n} + \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + \sum_{n=1}^{\infty} nc_n x^{n} - \sum_{n=0}^{\infty} c_n x^{n+1}$$

The first series begins with the x^2 term, the second series begins with the x^0 term, and the third and fourth series begin with the x^1 term, so the series aren't in phase. To put them in phase, we'll pull the x^0 and x^1 terms out of the second series, and we'll pull the x^1 term out of the third and fourth series.

© Krista King Math

$$\sum_{n=2}^{\infty} n(n-1)c_n x^n + 2(2-1)c_2 x^{2-2} + 3(3-1)c_3 x^{3-2} + \sum_{n=4}^{\infty} n(n-1)c_n x^{n-2}$$

$$+1c_1 x^1 + \sum_{n=2}^{\infty} nc_n x^n - c_0 x^{0+1} - \sum_{n=1}^{\infty} c_n x^{n+1}$$

$$2c_2 + (6c_3 + c_1 - c_0)x + \sum_{n=2}^{\infty} n(n-1)c_n x^n + \sum_{n=4}^{\infty} n(n-1)c_n x^{n-2}$$

$$+\sum_{n=2}^{\infty} nc_n x^n - \sum_{n=1}^{\infty} c_n x^{n+1}$$

The series are now in phase, but their indices don't match. To make the indices match, we'll make the substitution $k = n$ into the first and third series, make the substitution $k = n - 2$ and $n = k + 2$ into the second series, and make the substitution $k = n + 1$ and $n = k - 1$ into the fourth series.

$$2c_2 + (6c_3 + c_1 - c_0)x + \sum_{k=2}^{\infty} k(k-1)c_k x^k + \sum_{k=2}^{\infty} (k+2)(k+1)c_{k+2} x^k$$

$$+\sum_{k=2}^{\infty} kc_k x^k - \sum_{k=2}^{\infty} c_{k-1} x^k$$

The series are now in phase with matching indices, so we'll add them and then factor out x^k.

$$2c_2 + (6c_3 + c_1 - c_0)x + \sum_{k=2}^{\infty} k(k-1)c_k x^k + (k+2)(k+1)c_{k+2} x^k + kc_k x^k - c_{k-1} x^k$$

$$2c_2 + (6c_3 + c_1 - c_0)x + \sum_{k=2}^{\infty} [k(k-1)c_k + (k+2)(k+1)c_{k+2} + kc_k - c_{k-1}]x^k$$

© Krista King Math

$$2c_2 + (6c_3 + c_1 - c_0)x + \sum_{k=2}^{\infty} [(k+2)(k+1)c_{k+2} + k^2 c_k - c_{k-1}]x^k$$

6. Find the sum.

$$3x^2 \sum_{n=0}^{\infty} (n-2)(n-1)nc_n x^n - (x^2 - 2) \sum_{n=1}^{\infty} c_n x^{n+1} + (x+1) \sum_{n=2}^{\infty} n(n+1)c_n x^{n-1}$$

Solution:

Rewrite the second series by splitting it into two series.

$$3x^2 \sum_{n=0}^{\infty} (n-2)(n-1)nc_n x^n - x^2 \sum_{n=1}^{\infty} c_n x^{n+1} + 2 \sum_{n=1}^{\infty} c_n x^{n+1}$$

$$+(x+1) \sum_{n=2}^{\infty} n(n+1)c_n x^{n-1}$$

Rewrite the last series by splitting it into two series.

$$3x^2 \sum_{n=0}^{\infty} (n-2)(n-1)nc_n x^n - x^2 \sum_{n=1}^{\infty} c_n x^{n+1} + 2 \sum_{n=1}^{\infty} c_n x^{n+1}$$

$$+x \sum_{n=2}^{\infty} n(n+1)c_n x^{n-1} + \sum_{n=2}^{\infty} n(n+1)c_n x^{n-1}$$

Bring the coefficients into each series.

© Krista King Math

$$\sum_{n=0}^{\infty} 3(n-2)(n-1)nc_n x^{n+2} - \sum_{n=1}^{\infty} c_n x^{n+3} + \sum_{n=1}^{\infty} 2c_n x^{n+1}$$

$$+\sum_{n=2}^{\infty} n(n+1)c_n x^n + \sum_{n=2}^{\infty} n(n+1)c_n x^{n-1}$$

The first, third, and fourth series begin with the x^2 term, the second series begins with the x^4 term, and the fifth series begins with the x^1 term, so the series aren't in phase. To put them in phase, we'll pull all the x^0, x^1, x^2, and x^3 terms from each series.

$$3(0-2)(0-1)(0)c_0 x^{0+2} + 3(1-2)(1-1)(1)c_1 x^{1+2} + \sum_{n=2}^{\infty} 3(n-2)(n-1)nc_n x^{n+2}$$

$$-\sum_{n=1}^{\infty} c_n x^{n+3}$$

$$+2c_1 x^{1+1} + 2c_2 x^{2+1} + \sum_{n=3}^{\infty} 2c_n x^{n+1}$$

$$+2(2+1)c_2 x^2 + 3(3+1)c_3 x^3 + \sum_{n=4}^{\infty} n(n+1)c_n x^n$$

$$+2(2+1)c_2 x^{2-1} + 3(3+1)c_3 x^{3-1} + 4(4+1)c_4 x^{4-1} + \sum_{n=5}^{\infty} n(n+1)c_n x^{n-1}$$

$$\sum_{n=2}^{\infty} 3(n-2)(n-1)nc_n x^{n+2} - \sum_{n=1}^{\infty} c_n x^{n+3} + 2c_1 x^2 + 2c_2 x^3 + \sum_{n=3}^{\infty} 2c_n x^{n+1}$$

$$+6c_2 x^2 + 12c_3 x^3 + \sum_{n=4}^{\infty} n(n+1)c_n x^n$$

© Krista King Math

$$+6c_2x+12c_3x^2+20c_4x^3+\sum_{n=5}^{\infty}n(n+1)c_nx^{n-1}$$

$$6c_2x+(2c_1+6c_2+12c_3)x^2+(2c_2+12c_3+20c_4)x^3$$

$$+\sum_{n=2}^{\infty}3(n-2)(n-1)nc_nx^{n+2}-\sum_{n=1}^{\infty}c_nx^{n+3}+\sum_{n=3}^{\infty}2c_nx^{n+1}$$

$$+\sum_{n=4}^{\infty}n(n+1)c_nx^{n}+\sum_{n=5}^{\infty}n(n+1)c_nx^{n-1}$$

The series are now in phase, but their indices don't match. To make the indices match, we'll make the substitution $k=n+2$ and $n=k-2$ into the first series, make the substitution $k=n+3$ and $n=k-3$ into the second series, make the substitution $k=n+1$ and $n=k-1$ into the third series, make the substitution $k=n$ into the fourth series, and make the substitution $k=n-1$ and $n=k+1$ into the fifth series.

$$6c_2x+(2c_1+6c_2+12c_3)x^2+(2c_2+12c_3+20c_4)x^3$$

$$+\sum_{k=4}^{\infty}3(k-4)(k-3)(k-2)c_{k-2}x^k-\sum_{k=4}^{\infty}c_{k-3}x^k+\sum_{k=4}^{\infty}2c_{k-1}x^k$$

$$+\sum_{k=4}^{\infty}k(k+1)c_kx^k+\sum_{k=4}^{\infty}(k+1)(k+2)c_{k+1}x^k$$

The series are now in phase with matching indices, so we'll add them and then factor out x^k.

$$6c_2x+(2c_1+6c_2+12c_3)x^2+(2c_2+12c_3+20c_4)x^3$$

© Krista King Math

$$+\sum_{k=4}^{\infty}[3(k-4)(k-3)(k-2)c_{k-2}-c_{k-3}+2c_{k-1}+k(k+1)c_k$$

$$+(k+1)(k+2)c_{k+1}]x^k$$

$$6c_2x+(2c_1+6c_2+12c_3)x^2+(2c_2+12c_3+20c_4)x^3$$

$$+\sum_{k=4}^{\infty}[(k+1)(k+2)c_{k+1}+k(k+1)c_k+2c_{k-1}$$

$$+3(k-4)(k-3)(k-2)c_{k-2}-c_{k-3}]x^k$$

© Krista King Math

POWER SERIES SOLUTIONS

■ 1. Is $x_0 = 4$ an ordinary point of the differential equation?

$$\frac{1}{x-4}y'' + (x^2+2)y' + 3y = 0$$

Solution:

No, $x_0 = 4$ is a singular point of the differential equation, because the coefficient on y'' is undefined there. In other words, $1/(x-4)$ is undefined at $x_0 = 4$.

■ 2. Can we find a power series solution to the differential equation around $x_0 = -1$?

$$(x-1)y'' + xy' + (x^2+2)y = \sqrt{x-1}$$

Solution:

Even though all of the polynomial coefficients on the left side of the equation are all defined at $x_0 = -1$, the function on the right side of the equation, $g(x) = \sqrt{x-1}$, is undefined there, which means we won't be able to find a power series solution to the differential equation at that point.

© Krista King Math

■ 3. Find the minimum radius of convergence of a power series solution of the differential equation about the ordinary point $x_0 = -2$.

$$x^3y'' + (2x+1)y' - y = 0$$

Solution:

The singular points of the differential equation occur where $x^3 = 0$, or at $x = 0$. In the plane, we consider the ordinary point $x_0 = -2$ to occur at $(-2,0)$, and the singular point $x = 0$ to occur at $(0,0)$. The distance from the ordinary point $(-2,0)$ to the singular point $(0,0)$ is 2. So the minimum radius of convergence is

$$|x-(-2)| < 2$$

$$|x+2| < 2$$

■ 4. Find a power series solution to the differential equation, given $y(0) = 0$ and $y'(0) = 1$.

$$(x^2+1)y'' + 2xy' = 0$$

Solution:

© Krista King Math

We'll use the standard form for the first two derivatives of the power series solution,

$$y' = \sum_{n=1}^{\infty} c_n n x^{n-1} = c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 + \dots$$

$$y'' = \sum_{n=2}^{\infty} c_n n(n-1) x^{n-2} = 2c_2 + 6c_3 x + 12c_4 x^2 + \dots$$

to substitute into the differential equation.

$$(x^2+1)y'' + 2xy' = 0$$

$$(x^2+1)\sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} + 2x\sum_{n=1}^{\infty} c_n n x^{n-1} = 0$$

$$x^2\sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} + \sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} + 2x\sum_{n=1}^{\infty} c_n n x^{n-1} = 0$$

$$\sum_{n=2}^{\infty} c_n n(n-1)x^{n} + \sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} + \sum_{n=1}^{\infty} 2c_n n x^{n} = 0$$

The first series will start with the x^2 term, the second series will start with the x^0 term, and the third series will start with the x^1 term. We'll pull out all the x^0 and x^1 terms to put the series in phase.

$$\sum_{n=2}^{\infty} c_n n(n-1)x^{n} + c_2 2(2-1)x^{2-2} + c_3 3(3-1)x^{3-2} + \sum_{n=4}^{\infty} c_n n(n-1)x^{n-2}$$

$$+2c_1 1x^1 + \sum_{n=2}^{\infty} 2c_n n x^n = 0$$

© Krista King Math

$$2c_2 + (6c_3 + 2c_1)x + \sum_{n=2}^{\infty} c_n n(n-1)x^n + \sum_{n=4}^{\infty} c_n n(n-1)x^{n-2} + \sum_{n=2}^{\infty} 2c_n n x^n = 0$$

Now the series are in phase (they all begin with the x^2 term), so we just need to make their indices match. We'll substitute $k = n$ into the first and third series, and substitute $k = n - 2$ and $n = k + 2$ into the second series.

$$2c_2 + (6c_3 + 2c_1)x + \sum_{k=2}^{\infty} c_k k(k-1)x^k + \sum_{k=2}^{\infty} c_{k+2}(k+2)(k+1)x^k$$

$$+ \sum_{k=2}^{\infty} 2c_k k x^k = 0$$

Now that the series are in phase with matching indices, we can combine them and then factor out x^k.

$$2c_2 + (6c_3 + 2c_1)x + \sum_{k=2}^{\infty} c_k k(k-1)x^k + c_{k+2}(k+2)(k+1)x^k + 2c_k k x^k = 0$$

$$2c_2 + (6c_3 + 2c_1)x + \sum_{k=2}^{\infty} [c_k k(k-1) + c_{k+2}(k+2)(k+1) + 2c_k k]x^k = 0$$

$$2c_2 + (6c_3 + 2c_1)x + \sum_{k=2}^{\infty} [c_{k+2}(k+2)(k+1) + c_k k(k+1)]x^k = 0$$

Equating coefficients, we get

$$2c_2 = 0 \qquad k = 0$$

$$6c_3 + 2c_1 = 0 \qquad k = 1$$

$$c_{k+2}(k+2)(k+1) + c_k k(k+1) = 0 \qquad k = 2, 3, 4, \ldots$$

© Krista King Math

Solving the recurrence relation for the coefficient with the largest subscript, c_{k+2}, gives

$$c_2 = 0 \qquad k = 0$$

$$c_3 = -\frac{c_1}{3} \qquad k = 1$$

$$c_{k+2} = -\frac{c_k k}{k+2} \qquad k = 2, 3, 4, \ldots$$

Plugging in $k = 2, 3, 4, \ldots$, we start to see a pattern.

$$k = 0 \qquad c_2 = 0 \qquad k = 1 \qquad c_3 = -\frac{c_1}{3}$$

$$k = 2 \qquad c_4 = 0 \qquad k = 3 \qquad c_5 = \frac{c_1}{5}$$

$$k = 4 \qquad c_6 = 0 \qquad k = 5 \qquad c_7 = -\frac{c_1}{7}$$

$$\ldots \qquad \ldots$$

Plugging these coefficients into the expansion of the power series solution, we get

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \ldots$$

$$y = c_0 + c_1 x + 0x^2 - \frac{c_1}{3}x^3 + 0x^4 + \frac{c_1}{5}x^5 + 0x^6 - \frac{c_1}{7}x^7 + 0x^8 + \ldots$$

$$y = c_0 + c_1\left(x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \frac{1}{7}x^7 + \ldots\right)$$

So the general solution to the differential equation is

© Krista King Math

$$y = c_0 + c_1 \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$$

We've been told that $y(0) = 0$ and $y'(0) = 1$, so we can plug these into the formula for the series solution and its derivative to get

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \ldots$$

$$0 = c_0 + c_1(0) + c_2(0)^2 + c_3(0)^3 + \ldots$$

$$0 = c_0$$

and

$$y' = c_1 + 2c_2 x + 3c_3 x^2 + \ldots$$

$$1 = c_1 + 2c_2(0) + 3c_3(0)^2 + \ldots$$

$$1 = c_1$$

If we plug $c_0 = 0$ and $c_1 = 1$ back into the general solution, we get

$$y = 0 + 1 \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$$

$$y = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$$

We know that this is the power series representation of $\tan^{-1} x$,

$$\tan^{-1} x = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$$

© Krista King Math

which means we can write the general solution to the differential equation as

$$y = \tan^{-1} x$$

■ 5. Find a power series solution to the differential equation, given $y(0) = 1$ and $y'(0) = 0$.

$$y'' - xy' - y = 0$$

Solution:

We'll use the standard form for the power series solution and its derivatives,

$$y = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots$$

$$y' = \sum_{n=1}^{\infty} c_n n x^{n-1} = c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 + \dots$$

$$y'' = \sum_{n=2}^{\infty} c_n n(n-1) x^{n-2} = 2c_2 + 6c_3 x + 12c_4 x^2 + \dots$$

to substitute into the differential equation.

$$y'' - xy' - y = 0$$

© Krista King Math

$$\sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} - x\sum_{n=1}^{\infty} c_n n x^{n-1} - \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} - \sum_{n=1}^{\infty} c_n n x^{n} - \sum_{n=0}^{\infty} c_n x^n = 0$$

The first series will start with the $x^{2-2} = x^0$ term, the second series will start with the x^1 term, and the third series will start with the x^0 term. So we'll pull out the x^0 term from both the first and the third series in order to put the series in phase.

$$c_2(2)(2-1)x^{2-2} + \sum_{n=3}^{\infty} c_n n(n-1)x^{n-2} - \sum_{n=1}^{\infty} c_n n x^{n} - c_0 x^0 - \sum_{n=1}^{\infty} c_n x^n = 0$$

$$-c_0 + 2c_2 + \sum_{n=3}^{\infty} c_n n(n-1)x^{n-2} - \sum_{n=1}^{\infty} c_n n x^{n} - \sum_{n=1}^{\infty} c_n x^n = 0$$

Now the series are in phase (they all begin with the x^1 term), so we just need to make their indices match. We'll substitute $k = n - 2$ and $n = k + 2$ into the first series (the new index will be $k = 3 - 2 = 1$), and we'll substitute $k = n$ into the second and third series (their new indices will be $k = 1$).

$$-c_0 + 2c_2 + \sum_{k=1}^{\infty} c_{k+2}(k+2)(k+1)x^{k} - \sum_{k=1}^{\infty} c_k k x^{k} - \sum_{k=1}^{\infty} c_k x^k = 0$$

Now that the series are in phase with matching indices, we can combine them and then factor out x^k.

$$-c_0 + 2c_2 + \sum_{k=1}^{\infty} c_{k+2}(k+2)(k+1)x^{k} - c_k k x^{k} - c_k x^k = 0$$

© Krista King Math

$$-c_0 + 2c_2 + \sum_{k=1}^{\infty} (c_{k+2}(k+2)(k+1) - c_k k - c_k)x^k = 0$$

Equating coefficients, we get

$$-c_0 + 2c_2 = 0 \qquad k = 0$$

$$c_{k+2}(k+2)(k+1) - c_k k - c_k = 0 \qquad k = 1, 2, 3, 4, \ldots$$

Solving the recurrence relation for the coefficient with the largest subscript, c_{k+2}, gives

$$-c_0 + 2c_2 = 0 \qquad k = 0$$

$$c_{k+2} = \frac{c_k}{k+2} \qquad k = 1, 2, 3, 4, \ldots$$

Plugging in $k = 1, 2, 3, 4, \ldots$, we start to see a pattern.

$$k = 0 \qquad c_2 = \frac{c_0}{2} \qquad\qquad k = 1 \qquad c_3 = \frac{c_1}{3}$$

$$k = 2 \qquad c_4 = \frac{c_0}{2 \cdot 4} \qquad\qquad k = 3 \qquad c_5 = \frac{c_1}{3 \cdot 5}$$

$$k = 4 \qquad c_6 = \frac{c_0}{2 \cdot 4 \cdot 6} \qquad\qquad k = 5 \qquad c_7 = \frac{c_1}{3 \cdot 5 \cdot 7}$$

... ...

Plugging these coefficients into the expansion of the power series solution, we get

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \ldots$$

© Krista King Math

$$y = c_0 + c_1 x + \frac{c_0}{2}x^2 + \frac{c_1}{3}x^3 + \frac{c_0}{2 \cdot 4}x^4 + \frac{c_1}{3 \cdot 5}x^5 + \frac{c_0}{2 \cdot 4 \cdot 6}x^6 + \frac{c_1}{3 \cdot 5 \cdot 7}x^7 + \ldots$$

$$y = \left(c_0 + \frac{c_0}{2}x^2 + \frac{c_0}{2 \cdot 4}x^4 + \frac{c_0}{2 \cdot 4 \cdot 6}x^6 + \ldots \right)$$

$$+ \left(c_1 x + \frac{c_1}{3}x^3 + \frac{c_1}{3 \cdot 5}x^5 + \frac{c_1}{3 \cdot 5 \cdot 7}x^7 + \ldots \right)$$

$$y = c_0 \left(1 + \frac{1}{2}x^2 + \frac{1}{2 \cdot 4}x^4 + \frac{1}{2 \cdot 4 \cdot 6}x^6 + \ldots \right)$$

$$+ c_1 \left(x + \frac{1}{3}x^3 + \frac{1}{3 \cdot 5}x^5 + \frac{1}{3 \cdot 5 \cdot 7}x^7 + \ldots \right)$$

We could leave the solution this way, or we could try to rewrite each series in order to simplify the general solution.

$$y = c_0 \left(\frac{1}{2^0(0!)}x^{2(0)} + \frac{1}{2^1(1!)}x^{2(1)} + \frac{1}{2^2(2!)}x^{2(2)} + \frac{1}{2^3(3!)}x^{2(3)} + \ldots \right)$$

$$+ c_1 \left(\frac{2^0(0!)}{(2(0)+1)!}x^{2(0)+1} + \frac{2^1(1!)}{(2(1)+1)!}x^{2(1)+1} + \frac{2^2(2!)}{(2(2)+1)!}x^{2(2)+1} \right.$$

$$\left. + \frac{2^3(3!)}{(2(3)+1)!}x^{2(3)+1} + \ldots \right)$$

So the general solution to the differential equation is

$$y = c_0 \sum_{n=0}^{\infty} \frac{1}{2^n n!}x^{2n} + c_1 \sum_{n=0}^{\infty} \frac{2^n n!}{(2n+1)!}x^{2n+1}$$

© Krista King Math

We've been told that $y(0) = 1$ and $y'(0) = 0$, so we can plug these into the formula for the series solution and its derivative to get

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots$$

$$1 = c_0 + c_1(0) + c_2(0)^2 + c_3(0)^3 + \dots$$

$$1 = c_0$$

and

$$y' = c_1 + 2c_2 x + 3c_3 x^2 + \dots$$

$$0 = c_1 + 2c_2(0) + 3c_3(0)^2 + \dots$$

$$0 = c_1$$

If we plug $c_0 = 1$ and $c_1 = 0$ back into the general solution, we get

$$y = (1)\sum_{n=0}^{\infty} \frac{1}{2^n n!} x^{2n} + (0)\sum_{n=0}^{\infty} \frac{2^n n!}{(2n+1)!} x^{2n+1}$$

$$y = \sum_{n=0}^{\infty} \frac{1}{2^n n!} x^{2n}$$

We can leave the solution this way, or we can rewrite it as

$$y = \sum_{n=0}^{\infty} \left(\frac{x^2}{2}\right)^n \frac{1}{n!}$$

$$y = \sum_{n=0}^{\infty} \frac{\left(\frac{x^2}{2}\right)^n}{n!}$$

© Krista King Math

If we think back to our table of common power series, we remember that the series representation of e^x is given as

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$e^{\frac{x^2}{2}} = \sum_{n=0}^{\infty} \frac{\left(\frac{x^2}{2}\right)^n}{n!}$$

Therefore, we can write the general solution to the differential equation as

$$y = e^{\frac{x^2}{2}}$$

■ 6. Find a power series solution to the differential equation, given $y(0) = 0$ and $y'(0) = 1$.

$$y'' + x^2 y' + xy = 0$$

Solution:

We'll use the standard form for the power series solution and its derivatives,

$$y = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots$$

$$y' = \sum_{n=1}^{\infty} c_n n x^{n-1} = c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 + \dots$$

© Krista King Math



$$y'' = \sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} = 2c_2 + 6c_3x + 12c_4x^2 + \dots$$

to substitute into the differential equation.

$$y'' + x^2y' + xy = 0$$

$$\sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} + x^2 \sum_{n=1}^{\infty} c_n n x^{n-1} + x \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} + \sum_{n=1}^{\infty} c_n n x^{n+1} + \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

The first series will start with the $x^{2-2} = x^0$ term, the second series will start with the $x^{1+1} = x^2$ term, and the third series will start with the $x^{0+1} = x^1$ term. So we'll pull out the x^0 and x^1 terms from the first series, and we'll pull out the x^1 term from the third series. Doing so will put the series in phase.

$$c_2(2)(2-1)x^{2-2} + c_3(3)(3-1)x^{3-2} + \sum_{n=4}^{\infty} c_n n(n-1)x^{n-2}$$

$$+ \sum_{n=1}^{\infty} c_n n x^{n+1} + c_0 x^{0+1} + \sum_{n=1}^{\infty} c_n x^{n+1} = 0$$

$$2c_2 + (c_0 + 6c_3)x + \sum_{n=4}^{\infty} c_n n(n-1)x^{n-2} + \sum_{n=1}^{\infty} c_n n x^{n+1} + \sum_{n=1}^{\infty} c_n x^{n+1} = 0$$

Now the series are in phase (they all begin with the x^2 term), so we just need to make their indices match. We'll substitute $k = n - 2$ and $n = k + 2$ into the first series (the new index will be $k = 4 - 2 = 2$), and we'll substitute

© Krista King Math

$k = n + 1$ and $n = k - 1$ into the second and third series (their new indices will be $k = 2$).

$$2c_2 + (c_0 + 6c_3)x + \sum_{k=2}^{\infty} c_{k+2}(k+2)(k+1)x^k + \sum_{k=2}^{\infty} c_{k-1}(k-1)x^k + \sum_{k=2}^{\infty} c_{k-1}x^k = 0$$

Now that the series are in phase with matching indices, we can combine them and then factor out x^k.

$$2c_2 + (c_0 + 6c_3)x + \sum_{k=2}^{\infty} c_{k+2}(k+2)(k+1)x^k + c_{k-1}(k-1)x^k + c_{k-1}x^k = 0$$

$$2c_2 + (c_0 + 6c_3)x + \sum_{k=2}^{\infty} (c_{k+2}(k+2)(k+1) + c_{k-1}(k-1) + c_{k-1})x^k = 0$$

Equating coefficients, we get

$$2c_2 = 0 \qquad k = 0$$

$$c_0 + 6c_3 = 0 \qquad k = 1$$

$$c_{k+2}(k+2)(k+1) + c_{k-1}(k-1) + c_{k-1} = 0 \qquad k = 2, 3, 4, 5, \ldots$$

Solving the recurrence relation for the coefficient with the largest subscript, c_{k+2}, gives

$$c_2 = 0 \qquad k = 0$$

$$c_3 = -\frac{c_0}{6} \qquad k = 1$$

$$c_{k+2} = -\frac{kc_{k-1}}{(k+2)(k+1)} \qquad k = 2, 3, 4, 5, \ldots$$

© Krista King Math

Plugging in $k = 2, 3, 4, 5, \ldots$, we start to see a pattern.

For $k = 0, 3, 6, 9, \ldots$

$k = 0$ $\quad c_2 = 0$

$k = 3$ $\quad c_5 = 0$

$k = 6$ $\quad c_8 = 0$

$k = 9$ $\quad c_{11} = 0$

For $k = 1, 4, 7, 10, \ldots$

$k = 1$ $\quad c_3 = -\dfrac{c_0}{6}$

$k = 4$ $\quad c_6 = \dfrac{4c_0}{(6)(6)(5)}$

$k = 7$ $\quad c_9 = -\dfrac{7(4)c_0}{(9)(8)(6)(6)(5)}$

$k = 10$ $\quad c_{12} = \dfrac{10(7)(4)c_0}{(12)(11)(9)(8)(6)(6)(5)}$

For $k = 2, 5, 8, 11, \ldots$

$k = 2$ $\quad c_4 = -\dfrac{2c_1}{(4)(3)}$

$k = 5$ $\quad c_7 = \dfrac{5(2)c_1}{(7)(6)(4)(3)}$

© Krista King Math



$$k = 8 \qquad c_{10} = -\frac{8(5)(2)c_1}{(10)(9)(7)(6)(4)(3)}$$

$$k = 11 \qquad c_{13} = \frac{11(8)(5)(2)c_1}{(13)(12)(10)(9)(7)(6)(4)(3)}$$

We've been told that $y(0) = 0$ and $y'(0) = 1$, so we can plug these into the formula for the series solution and its derivative to get

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + \dots$$

$$0 = c_0 + c_1(0) + c_2(0)^2 + c_3(0)^3 + \dots$$

$$0 = c_0$$

and

$$y' = c_1 + 2c_2x + 3c_3x^2 + \dots$$

$$1 = c_1 + 2c_2(0) + 3c_3(0)^2 + \dots$$

$$1 = c_1$$

Which means our coefficients become

$$k = 0 \quad c_2 = 0 \qquad k = 1 \quad c_3 = 0 \qquad k = 2 \quad c_4 = -\frac{2}{(4)(3)}$$

$$k = 3 \quad c_5 = 0 \qquad k = 4 \quad c_6 = 0 \qquad k = 5 \quad c_7 = \frac{5(2)}{(7)(6)(4)(3)}$$

$$k = 6 \quad c_8 = 0 \qquad k = 7 \quad c_9 = 0 \qquad k = 8 \quad c_{10} = -\frac{8(5)(2)}{(10)(9)(7)(6)(4)(3)}$$

© Krista King Math

$$k = 9 \quad c_{11} = 0 \qquad k = 10 \quad c_{12} = 0 \qquad k = 11 \quad c_{13} = \frac{11(8)(5)(2)}{(13)(12)(10)(9)(7)(6)(4)(3)}$$

Plugging these coefficients into the expansion of the power series solution, we get

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + c_6x^6 + c_7x^7 + c_8x^8 + \dots$$

$$y = 0 + x + 0x^2 - 0x^3 - \frac{2}{(4)(3)}x^4 + 0x^5 + 0x^6 + \frac{5(2)}{(7)(6)(4)(3)}x^7$$

$$+0x^8 - 0x^9 - \frac{8(5)(2)}{(10)(9)(7)(6)(4)(3)}x^{10} + \dots$$

$$y = x - \frac{2}{(4)(3)}x^4 + \frac{5(2)}{(7)(6)(4)(3)}x^7 - \frac{8(5)(2)}{(10)(9)(7)(6)(4)(3)}x^{10}$$

$$+\frac{11(8)(5)(2)}{(13)(12)(10)(9)(7)(6)(4)(3)}x^{13} - \dots$$

We could leave the solution this way, or we could try to rewrite each series in order to simplify the general solution.

$$y = x - \frac{2(2)}{4!}x^4 + \frac{5(5)(2)(2)}{7!}x^7 - \frac{8(8)(5)(5)(2)(2)}{10!}x^{10}$$

$$+\frac{11(11)(8)(8)(5)(5)(2)(2)}{13!}x^{13} - \dots$$

$$y = x - \frac{(2)^2}{4!}x^4 + \frac{(2 \cdot 5)^2}{7!}x^7 - \frac{(2 \cdot 5 \cdot 8)^2}{10!}x^{10} + \frac{(2 \cdot 5 \cdot 8 \cdot 11)^2}{13!}x^{13} - \dots$$

So the general solution to the differential equation is

© Krista King Math

$$y = x + \sum_{n=1}^{\infty} (-1)^n \frac{2^2 \cdot 5^2 \cdot 8^2 \cdot 11^2 \cdot \ldots \cdot (3n-1)^2 x^{3n+1}}{(3n+1)!}$$

© Krista King Math

NONPOLYNOMIAL COEFFICIENTS

■ 1. Solve the differential equation around $x_0 = 0$.

$$y'' + \frac{1}{1-x}y' = 0$$

Solution:

We know that the power series representations of the first two derivatives y' and y'' are

$$y' = \sum_{n=1}^{\infty} c_n n x^{n-1} = c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 + 5c_5 x^4 + \dots$$

$$y'' = \sum_{n=2}^{\infty} c_n n(n-1) x^{n-2} = 2c_2 + 6c_3 x + 12c_4 x^2 + 20c_5 x^3 + 30c_6 x^4 + \dots$$

and that the power series representation of the coefficient function on y' is

$$\frac{1}{1-x} = \sum_{n=0} x^n = 1 + x + x^2 + x^3 + x^4 + \dots$$

Plugging all of these values into the original differential equation gives

$$\sum_{n=2} n(n-1)c_n x^{n-2} + (1 + x + x^2 + x^3 + x^4 + \dots) \sum_{n=1} n c_n x^{n-1} = 0$$

Replace each power series with its expanded version.

© Krista King Math

$$(2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 + 30c_6x^4 + \dots)$$

$$+(1 + x + x^2 + x^3 + x^4 + \dots)(c_1 + 2c_2x + 3c_3x^2 + 4c_4x^3 + 5c_5x^4 + \dots) = 0$$

$$2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 + 30c_6x^4 + \dots$$

$$+c_1 + 2c_2x + 3c_3x^2 + 4c_4x^3 + 5c_5x^4 + \dots$$

$$+c_1x + 2c_2x^2 + 3c_3x^3 + 4c_4x^4 + 5c_5x^5 + \dots$$

$$+c_1x^2 + 2c_2x^3 + 3c_3x^4 + 4c_4x^5 + 5c_5x^6 + \dots$$

$$+c_1x^3 + 2c_2x^4 + 3c_3x^5 + 4c_4x^6 + 5c_5x^7 + \dots$$

$$+c_1x^4 + 2c_2x^5 + 3c_3x^6 + 4c_4x^7 + 5c_5x^8 + \dots = 0$$

$$(2c_2 + c_1) + (6c_3 + 2c_2 + c_1)x + (12c_4 + 3c_3 + 2c_2 + c_1)x^2$$

$$+(20c_5 + 4c_4 + 3c_3 + 2c_2 + c_1)x^3 + \dots = 0$$

This lets us build a system of equations.

$$2c_2 + c_1 = 0$$

$$6c_3 + 2c_2 + c_1 = 0$$

$$12c_4 + 3c_3 + 2c_2 + c_1 = 0$$

$$20c_5 + 4c_4 + 3c_3 + 2c_2 + c_1 = 0$$

Solving this system, we get $c_2 = -c_1/2$, $c_3 = 0$, $c_4 = 0$, and $c_5 = 0$. Therefore, the power series representation of the solution to the differential equation will be

© Krista King Math

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + \dots$$

$$y = c_0 + c_1x - \frac{c_1}{2}x^2$$

■ 2. Find the first six terms of the power series solutions of the differential equation around the ordinary point $x_0 = 0$.

$$y'' - e^x y = 0$$

Solution:

We know $x_0 = 0$ is an ordinary point because e^x is analytic there. And we know that the Maclaurin series representation of e^x is

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots$$

Plugging this and the power series representations of y and y'',

$$y = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + \dots$$

$$y'' = \sum_{n=2}^{\infty} c_n n(n-1)x^{n-2} = 2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 + 30c_6x^4 + \dots$$

into the original differential equation, we get

© Krista King Math

$$\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} - \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots\right)\sum_{n=0}^{\infty} c_n x^n = 0$$

Now we'll just expand each series through its first few terms.

$$(2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 + 30c_6x^4 + \dots)$$

$$-\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots\right)(c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + \dots) = 0$$

$$2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 + 30c_6x^4 + \dots$$

$$-c_0 - c_1x - c_2x^2 - c_3x^3 - c_4x^4 - \dots$$

$$-c_0x - c_1x^2 - c_2x^3 - c_3x^4 - c_4x^5 - \dots$$

$$-c_0\frac{x^2}{2!} - c_1\frac{x^3}{2!} - c_2\frac{x^4}{2!} - c_3\frac{x^5}{2!} - c_4\frac{x^6}{2!} - \dots$$

$$-c_0\frac{x^3}{3!} - c_1\frac{x^4}{3!} - c_2\frac{x^5}{3!} - c_3\frac{x^6}{3!} - c_4\frac{x^7}{3!} - \dots$$

$$-c_0\frac{x^4}{4!} - c_1\frac{x^5}{4!} - c_2\frac{x^6}{4!} - c_3\frac{x^7}{4!} - c_4\frac{x^8}{4!} - \dots$$

$$-c_0\frac{x^5}{5!} - c_1\frac{x^6}{5!} - c_2\frac{x^7}{5!} - c_3\frac{x^8}{5!} - c_4\frac{x^9}{5!} - \dots = 0$$

$$(2c_2 - c_0) + (6c_3 - c_1 - c_0)x + \left(12c_4 - c_2 - c_1 - \frac{c_0}{2!}\right)x^2$$

$$+\left(20c_5 - c_3 - c_2 - \frac{c_1}{2!} - \frac{c_0}{3!}\right)x^3 + \dots = 0$$

© Krista King Math

This lets us build a system of equations.

$$2c_2 - c_0 = 0$$

$$6c_3 - c_1 - c_0 = 0$$

$$12c_4 - c_2 - c_1 - \frac{c_0}{2!} = 0$$

$$20c_5 - c_3 - c_2 - \frac{c_1}{2!} - \frac{c_0}{3!} = 0$$

This system simplifies to

$$c_2 = \frac{c_0}{2}$$

$$c_3 = \frac{c_0 + c_1}{6}$$

$$c_4 = \frac{c_0 + c_1}{12}$$

$$c_5 = \frac{5c_0 + 4c_1}{120}$$

Therefore, the power series representation of the solution to the differential equation will be

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + \dots$$

$$y = c_0 + c_1x + \frac{c_0}{2}x^2 + \frac{c_0 + c_1}{6}x^3 + \frac{c_0 + c_1}{12}x^4 + \frac{5c_0 + 4c_1}{120}x^5 + \dots$$

$$y = c_0\left(1 + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{12}x^4 + \dots\right) + c_1\left(x + \frac{1}{6}x^3 + \frac{1}{12}x^4 + \frac{1}{30}x^5 + \dots\right)$$

© Krista King Math

■ 3. Find the first five terms of the power series solutions of the differential equation around the ordinary point $x_0 = 0$.

$$y' + e^{3x}y = 0$$

Solution:

We know $x_0 = 0$ is an ordinary point because e^{3x} is analytic there. And we know that the Maclaurin series representation of e^{3x} is

$$e^{3x} = \sum_{n=0}^{\infty} \frac{3^n x^n}{n!} = 1 + 3x + \frac{9x^2}{2!} + \frac{27x^3}{3!} + \frac{81x^4}{4!} + \frac{243x^5}{5!} + \dots$$

Plugging this and the power series representations of y and y',

$$y = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \dots$$

$$y' = \sum_{n=1}^{\infty} c_n n x^{n-1} = c_1 + 2c_2 x + 3c_3 x^2 + 4c_4 x^3 + 5c_5 x^4 + \dots$$

into the original differential equation, we get

$$\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + \left(1 + 3x + \frac{9x^2}{2!} + \frac{27x^3}{3!} + \frac{81x^4}{4!} + \frac{243x^5}{5!} + \dots\right)\sum_{n=0}^{\infty} c_n x^n = 0$$

Now we'll just expand each series through its first few terms.

© Krista King Math

$$(c_1 + 2c_2x + 3c_3x^2 + 4c_4x^3 + 5c_5x^4 + \dots)$$

$$+\left(1 + 3x + \frac{9x^2}{2!} + \frac{27x^3}{3!} + \frac{81x^4}{4!} + \frac{243x^5}{5!} + \dots\right)$$

$$(c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + \dots) = 0$$

$$c_1 + 2c_2x + 3c_3x^2 + 4c_4x^3 + 5c_5x^4 + \dots$$

$$+c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + \dots$$

$$+3c_0x + 3c_1x^2 + 3c_2x^3 + 3c_3x^4 + 3c_4x^5 + \dots$$

$$+\frac{9c_0}{2!}x^2 + \frac{9c_1}{2!}x^3 + \frac{9c_2}{2!}x^4 + \frac{9c_3}{2!}x^5 + \frac{9c_4}{2!}x^6 + \dots$$

$$+\frac{27c_0}{3!}x^3 + \frac{27c_1}{3!}x^4 + \frac{27c_2}{3!}x^5 + \frac{27c_3}{3!}x^6 + \frac{27c_4}{3!}x^7 + \dots$$

$$+\frac{81c_0}{4!}x^4 + \frac{81c_1}{4!}x^5 + \frac{81c_2}{4!}x^6 + \frac{81c_3}{4!}x^7 + \frac{81c_4}{4!}x^8 + \dots$$

$$+\frac{243c_0}{5!}x^5 + \frac{243c_1}{5!}x^6 + \frac{243c_2}{5!}x^7 + \frac{243c_3}{5!}x^8 + \frac{243c_4}{5!}x^9 + \dots = 0$$

$$(c_1 + c_0) + (2c_2 + c_1 + 3c_0)x + \left(3c_3 + c_2 + 3c_1 + \frac{9c_0}{2!}\right)x^2$$

$$+\left(4c_4 + c_3 + 3c_2 + \frac{9c_1}{2!} + \frac{27c_0}{3!}\right)x^3 + \dots = 0$$

This lets us build a system of equations.

$$c_1 + c_0 = 0$$

© Krista King Math

$$2c_2 + c_1 + 3c_0 = 0$$

$$3c_3 + c_2 + 3c_1 + \frac{9c_0}{2!} = 0$$

$$4c_4 + c_3 + 3c_2 + \frac{9c_1}{2!} + \frac{27c_0}{3!} = 0$$

This system simplifies to

$$c_1 = -c_0$$

$$c_2 = -c_0$$

$$c_3 = -\frac{c_0}{6}$$

$$c_4 = \frac{19c_0}{24}$$

Therefore, the power series representation of the solution to the differential equation will be

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + \dots$$

$$y(x) = c_0 - c_0x - c_0x^2 - \frac{c_0}{6}x^3 + \frac{19c_0}{24}x^4 + \dots$$

■ 4. Find the first eight terms of the power series solution of the differential equation around the ordinary point $x_0 = 0$.

$$y'' + y\cos x = 0$$

© Krista King Math

Solution:

We know $x_0 = 0$ is an ordinary point because $\cos x$ is analytic there. And we know that the Maclaurin series representation of $\cos x$ is

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

Plugging this and the power series representations of y and y'',

$$y = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \dots$$

$$y'' = \sum_{n=2}^{\infty} c_n n(n-1) x^{n-2} = 2c_2 + 6c_3 x + 12c_4 x^2 + 20c_5 x^3 + 30c_6 x^4 + \dots$$

into the original differential equation, we get

$$\sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} + \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots\right) \sum_{n=0}^{\infty} c_n x^n = 0$$

Now we'll just expand each series through its first few terms.

$$(2c_2 + 6c_3 x + 12c_4 x^2 + 20c_5 x^3 + \dots)$$

$$+ \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots\right)(c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots) = 0$$

$$(2c_2 + 6c_3 x + 12c_4 x^2 + 20c_5 x^3 + \dots)$$

© Krista King Math

$$+(c_0 + c_1x + c_2x^2 + c_3x^3 + \dots)$$

$$-\left(\frac{c_0}{2!}x^2 + \frac{c_1}{2!}x^3 + \frac{c_2}{2!}x^4 + \frac{c_3}{2!}x^5 + \dots\right)$$

$$+\left(\frac{c_0}{4!}x^4 + \frac{c_1}{4!}x^5 + \frac{c_2}{4!}x^6 + \frac{c_3}{4!}x^7 + \dots\right)$$

$$-\left(\frac{c_0}{6!}x^6 + \frac{c_1}{6!}x^7 + \frac{c_2}{6!}x^8 + \frac{c_3}{6!}x^9 + \dots\right) - \dots = 0$$

$$(2c_2 + c_0) + (6c_3 + c_1)x + \left(12c_4 + c_2 - \frac{c_0}{2!}\right)x^2$$

$$+\left(20c_5 + c_3 - \frac{c_1}{2!}\right)x^3 + \left(30c_6 + c_4 - \frac{c_2}{2!} + \frac{c_0}{4!}\right)x^4$$

$$+\left(42c_7 + c_5 - \frac{c_3}{2!} + \frac{c_1}{4!}\right)x^5 + \dots = 0$$

This lets us build a system of equations.

$$2c_2 + c_0 = 0$$

$$6c_3 + c_1 = 0$$

$$12c_4 + c_2 - \frac{c_0}{2!} = 0$$

$$20c_5 + c_3 - \frac{c_1}{2!} = 0$$

$$30c_6 + c_4 - \frac{c_2}{2!} + \frac{c_0}{4!} = 0$$

© Krista King Math

$$42c_7 + c_5 - \frac{c_3}{2!} + \frac{c_1}{4!} = 0$$

This system simplifies to

$$c_2 = -\frac{c_0}{2} \qquad c_3 = -\frac{c_1}{6}$$

$$c_4 = \frac{c_0}{12} \qquad c_5 = \frac{c_1}{30}$$

$$c_6 = -\frac{c_0}{80} \qquad c_7 = -\frac{19c_1}{5{,}040}$$

Therefore, the power series representation of the solution to the differential equation will be

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + \dots$$

$$y(x) = c_0 + c_1x - \frac{c_0}{2}x^2 - \frac{c_1}{6}x^3 + \frac{c_0}{12}x^4 + \frac{c_1}{30}x^5 - \frac{c_0}{80}x^6 + \frac{19c_1}{5{,}040}x^7 + \dots$$

■ 5. Find the first six terms of the power series solution of the differential equation around the ordinary point $x_0 = 0$.

$$y'' - \ln(1 + 5x)y = 0$$

Solution:

We know $x_0 = 0$ is an ordinary point because $\ln(1 + 5x)$ is analytic there. And we know that the Maclaurin series representation of $\ln(1 + 5x)$ is

© Krista King Math

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \dots$$

$$\ln(1+5x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} (5x)^n = 5x - \frac{1}{2}(5x)^2 + \frac{1}{3}(5x)^3 - \frac{1}{4}(5x)^4 + \dots$$

$$\ln(1+5x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}5^n}{n} x^n = 5x - \frac{25}{2}x^2 + \frac{125}{3}x^3 - \frac{625}{4}x^4 + \dots$$

Plugging this and the power series representations of y and y'',

$$y = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \dots$$

$$y'' = \sum_{n=2}^{\infty} c_n n(n-1) x^{n-2} = 2c_2 + 6c_3 x + 12c_4 x^2 + 20c_5 x^3 + 30c_6 x^4 + \dots$$

into the original differential equation, we get

$$\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} - \left(5x - \frac{25}{2}x^2 + \frac{125}{3}x^3 - \frac{625}{4}x^4 + \dots\right) \sum_{n=0}^{\infty} c_n x^n = 0$$

Now we'll just expand each series through its first few terms.

$$(2c_2 + 6c_3 x + 12c_4 x^2 + 20c_5 x^3 + \dots)$$

$$-\left(5x - \frac{25}{2}x^2 + \frac{125}{3}x^3 - \frac{625}{4}x^4 + \dots\right)(c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots) = 0$$

$$(2c_2 + 6c_3 x + 12c_4 x^2 + 20c_5 x^3 + \dots)$$

$$-5x(c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots)$$

© Krista King Math

$$+\frac{25}{2}x^2(c_0+c_1x+c_2x^2+c_3x^3+\dots)$$

$$-\frac{125}{3}x^3(c_0+c_1x+c_2x^2+c_3x^3+\dots)-\dots=0$$

$$(2c_2+6c_3x+12c_4x^2+20c_5x^3+\dots)$$

$$-(5c_0x+5c_1x^2+5c_2x^3+5c_3x^4+\dots)$$

$$+\left(\frac{25c_0}{2}x^2+\frac{25c_1}{2}x^3+\frac{25c_2}{2}x^4+\frac{25c_3}{2}x^5+\dots\right)$$

$$-\left(\frac{125c_0}{3}x^3+\frac{125c_1}{3}x^4+\frac{125c_2}{3}x^5+\frac{125c_3}{3}x^6+\dots\right)-\dots=0$$

$$2c_2+(6c_3-5c_0)x+\left(12c_4-5c_1+\frac{25c_0}{2}\right)x^2$$

$$+\left(20c_5-5c_2+\frac{25c_1}{2}-\frac{125c_0}{3}\right)x^3+\dots=0$$

This lets us build a system of equations.

$$2c_2=0$$

$$6c_3-5c_0=0$$

$$12c_4-5c_1+\frac{25c_0}{2}=0$$

$$20c_5-5c_2+\frac{25c_1}{2}-\frac{125c_0}{3}=0$$

This system simplifies to

© Krista King Math

$$c_2 = 0$$

$$c_3 = \frac{5c_0}{6}$$

$$c_4 = \frac{10c_1 - 25c_0}{24}$$

$$c_5 = \frac{50c_0 - 15c_1}{24}$$

Therefore, the power series representation of the solution to the differential equation will be

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + \dots$$

$$y = c_0 + c_1x + \frac{5c_0}{6}x^3 + \frac{10c_1 - 25c_0}{24}x^4 + \frac{50c_0 - 15c_1}{24}x^5 + \dots$$

■ 6. Find the first four terms of the power series solution of the differential equation around the ordinary point $x_0 = 2$.

$$y'' + \cosh(x - 2)y = 0$$

Solution:

We know $x_0 = 0$ is an ordinary point because $\cosh(x - 2)$ is analytic there. And we know that the Maclaurin series representation of $\cosh(x - 2)$ is

© Krista King Math



$$\cosh x = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \frac{x^8}{8!} + \dots$$

$$\cosh(x-2) = \sum_{n=0}^{\infty} \frac{(x-2)^{2n}}{(2n)!} = 1 + \frac{(x-2)^2}{2!} + \frac{(x-2)^4}{4!} + \frac{(x-2)^6}{6!} + \frac{(x-2)^8}{8!} + \dots$$

Plugging this and the power series representations of $y(x-2)$ and $y''(x-2)$,

$$y(x-2) = \sum_{n=0}^{\infty} c_n (x-2)^n$$

$$= c_0 + c_1(x-2) + c_2(x-2)^2 + c_3(x-2)^3 + c_4(x-2)^4 + \dots$$

$$y''(x-2) = \sum_{n=2}^{\infty} c_n n(n-1)(x-2)^{n-2}$$

$$= 2c_2 + 6c_3(x-2) + 12c_4(x-2)^2 + 20c_5(x-2)^3 + 30c_6(x-2)^4 + \dots$$

into the original differential equation, we get

$$\sum_{n=2}^{\infty} n(n-1)c_n(x-2)^{n-2} + \sum_{n=0}^{\infty} \frac{(x-2)^{2n}}{(2n)!} \sum_{n=0}^{\infty} c_n(x-2)^n = 0$$

Now we'll just expand each series through its first few terms.

$$(2c_2 + 6c_3(x-2) + 12c_4(x-2)^2 + 20c_5(x-2)^3 + \dots)$$

$$+\left(1 + \frac{(x-2)^2}{2!} + \frac{(x-2)^4}{4!} + \frac{(x-2)^6}{6!} + \dots\right)$$

$$(c_0 + c_1(x-2) + c_2(x-2)^2 + c_3(x-2)^3 + \dots) + \dots = 0$$

© Krista King Math

$$(2c_2 + 6c_3(x-2) + 12c_4(x-2)^2 + 20c_5(x-2)^3 + \dots)$$

$$+(c_0 + c_1(x-2) + c_2(x-2)^2 + c_3(x-2)^3 + \dots)$$

$$+\frac{(x-2)^2}{2!}(c_0 + c_1(x-2) + c_2(x-2)^2 + c_3(x-2)^3 + \dots)$$

$$+\frac{(x-2)^4}{4!}(c_0 + c_1(x-2) + c_2(x-2)^2 + c_3(x-2)^3 + \dots) + \dots = 0$$

$$(2c_2 + 6c_3(x-2) + 12c_4(x-2)^2 + 20c_5(x-2)^3 + \dots)$$

$$+(c_0 + c_1(x-2) + c_2(x-2)^2 + c_3(x-2)^3 + \dots)$$

$$+\left(\frac{c_0}{2!}(x-2)^2 + \frac{c_1}{2!}(x-2)^3 + \frac{c_2}{2!}(x-2)^4 + \frac{c_3}{2!}(x-2)^5 + \dots\right)$$

$$+\left(\frac{c_0}{4!}(x-2)^4 + \frac{c_1}{4!}(x-2)^5 + \frac{c_2}{4!}(x-2)^6 + \frac{c_3}{4!}(x-2)^7 + \dots\right) + \dots = 0$$

$$(2c_2 + c_0) + (6c_3 + c_1)(x-2) + \left(12c_4 + c_2 + \frac{c_0}{2!}\right)(x-2)^2$$

$$+\left(20c_5 + c_3 + \frac{c_1}{2!}\right)(x-2)^3 + \dots = 0$$

This lets us build a system of equations.

$$2c_2 + c_0 = 0$$

$$6c_3 + c_1 = 0$$

$$12c_4 + c_2 + \frac{c_0}{2!} = 0$$

© Krista King Math

$$20c_5 + c_3 + \frac{c_1}{2!} = 0$$

This system simplifies to

$$c_2 = -\frac{c_0}{2}$$

$$c_3 = -\frac{c_1}{6}$$

$$c_4 = 0$$

Therefore, the power series representation of the solution to the differential equation will be

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 + \dots$$

$$y = c_0 + c_1(x-2) - \frac{c_0}{2}(x-2)^2 - \frac{c_1}{6}(x-2)^3 + \dots$$

© Krista King Math



SINGULAR POINTS AND FROBENIUS' THEOREM

■ 1. Determine the regularity of the singular point $x_0 = 0$ of the differential equation, use the method of Frobenius to build any solution(s) around that point, then find the general solution.

$$2x^2y'' + x(1 - 2x)y' + 4xy = 0$$

Solution:

Matching this differential equation to the standard form $p(x)y'' + q(x)y' + r(x)y = 0$, we can identify

$$p(x) = 2x^2$$

$$q(x) = x(1 - 2x)$$

$$r(x) = 4x$$

Using these three functions to calculate $Q(x)$ and $R(x)$ gives

$$Q(x) = (x - x_0)\frac{q(x)}{p(x)} = x\frac{x(1 - 2x)}{2x^2} = \frac{1 - 2x}{2}$$

$$R(x) = (x - x_0)^2\frac{r(x)}{p(x)} = x^2\frac{4x}{2x^2} = 2x$$

Because both denominators simplify to constants, we can see that both $Q(x)$ and $R(x)$ are analytic about $x_0 = 0$, so $x_0 = 0$ is a regular singular point.

© Krista King Math

Now we'll use

$$y = \sum_{n=0}^{\infty} c_n x^{n+r}$$

$$y' = \sum_{n=0}^{\infty} (n+r)c_n x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1)c_n x^{n+r-2}$$

to make substitutions into the differential equation.

$$2x^2y'' + x(1-2x)y' + 4xy = 0$$

$$2x^2 \sum_{n=0}^{\infty} (n+r)(n+r-1)c_n x^{n+r-2} + x(1-2x) \sum_{n=0}^{\infty} (n+r)c_n x^{n+r-1}$$

$$+4x \sum_{n=0}^{\infty} c_n x^{n+r} = 0$$

$$\sum_{n=0}^{\infty} 2(n+r)(n+r-1)c_n x^{n+r} + \sum_{n=0}^{\infty} (n+r)c_n x^{n+r} - \sum_{n=0}^{\infty} 2(n+r)c_n x^{n+r+1}$$

$$+ \sum_{n=0}^{\infty} 4c_n x^{n+r+1} = 0$$

Combine terms with equivalent powers of x.

$$\sum_{n=0}^{\infty} [4 - 2(n+r)]c_n x^{n+r+1} + \sum_{n=0}^{\infty} [2(n+r)(n+r-1) + (n+r)]c_n x^{n+r} = 0$$

© Krista King Math

$$\sum_{n=0}^{\infty}[4-2(n+r)]c_n x^{n+r+1}+\sum_{n=0}^{\infty}(n+r)(2n+2r-1)c_n x^{n+r}=0$$

$$x^r\left[\sum_{n=0}^{\infty}[4-2(n+r)]c_n x^{n+1}+\sum_{n=0}^{\infty}(n+r)(2n+2r-1)c_n x^{n}\right]=0$$

The first series starts with the x^1 term, while the second series starts with the x^0 term, so we'll pull the x^0 term out of the second series.

$$x^r\left[\sum_{n=0}^{\infty}[4-2(n+r)]c_n x^{n+1}+r(2r-1)c_0+\sum_{n=1}^{\infty}(n+r)(2n+2r-1)c_n x^{n}\right]=0$$

Now that the series are in phase, we'll substitute $k=n+1$ and $n=k-1$ into the first series, and $k=n$ into the second series.

$$x^r\left[r(2r-1)c_0+\sum_{k=1}^{\infty}[4-2(k+r-1)]c_{k-1}x^{k}+\sum_{k=1}^{\infty}(k+r)(2k+2r-1)c_k x^{k}\right]=0$$

Now that the series are in phase with matching indices, combine them.

$$x^r\left[r(2r-1)c_0+\sum_{k=1}^{\infty}[(6-2k-2r)c_{k-1}+(k+r)(2k+2r-1)c_k]x^{k}\right]=0$$

This equation gives

$$r(2r-1)c_0=0 \qquad k=0$$

$$(6-2k-2r)c_{k-1}+(k+r)(2k+2r-1)c_k=0 \qquad k=1,\ 2,\ 3,\ \ldots$$

or

© Krista King Math

$$r(2r-1)=0 \qquad k=0$$

$$c_k=-\frac{(6-2k-2r)c_{k-1}}{(k+r)(2k+2r-1)} \qquad k=1,\,2,\,3,\,\ldots$$

The indicial equation gives us $r_1=1/2$ and $r_2=0$. Substituting these indicial roots into the recurrence relation, we get

	For $r_1=1/2$	For $r_2=0$
	$c_k=-\frac{(5-2k)c_{k-1}}{k(2k+1)}$	$c_k=-\frac{(6-2k)c_{k-1}}{k(2k-1)}$
$k=1$	$c_1=-c_0$	$c_1=-4c_0$
$k=2$	$c_2=\frac{1}{10}c_0$	$c_2=\frac{4}{3}c_0$
$k=3$	$c_3=\frac{1}{210}c_0$	$c_3=0$
...	...	...

Forming these coefficients into series around $x_0=0$ gives,

$$c_0(x-x_0)^0+c_1(x-x_0)^1+c_2(x-x_0)^2+c_3(x-x_0)^3+\ldots$$

$$c_0x^0-c_0x^1+\frac{1}{10}c_0x^2+\frac{1}{210}c_0x^3+\ldots$$

$$c_0\left(1-x+\frac{1}{10}x^2+\frac{1}{210}x^3+\ldots\right)$$

and

© Krista King Math

$$c_0(x-x_0)^0 + c_1(x-x_0)^1 + c_2(x-x_0)^2 + c_3(x-x_0)^3 + \dots$$

$$c_0x^0 - 4c_0x^1 + \frac{4}{3}c_0x^2 + \dots$$

$$c_0\left(1 - 4x + \frac{4}{3}x^2 + \dots\right)$$

So the series solutions are

$$y_1(x) = c_0x^{\frac{1}{2}}\left(1 - x + \frac{1}{10}x^2 + \frac{1}{210}x^3 + \dots\right)$$

$$y_2(x) = c_0x^0\left(1 - 4x + \frac{4}{3}x^2 + \dots\right)$$

and the general solution is

$$y(x) = C_1y_1(x) + C_2y_2(x)$$

$$y(x) = C_1c_0x^{\frac{1}{2}}\left(1 - x + \frac{1}{10}x^2 + \frac{1}{210}x^3 + \dots\right) + C_2c_0\left(1 - 4x + \frac{4}{3}x^2 + \dots\right)$$

$$y(x) = C_1x^{\frac{1}{2}}\left(1 - x + \frac{1}{10}x^2 + \dots\right) + C_2\left(1 - 4x + \frac{4}{3}x^2 + \dots\right)$$

■ 2. Determine the regularity of the singular point $x_0 = 0$ of the differential equation, use the method of Frobenius to build any solution(s) around that point, then find the general solution.

$$4xy'' + 2y' + y = 0$$

© Krista King Math

Solution:

Matching this differential equation to the standard form $p(x)y'' + q(x)y' + r(x)y = 0$, we can identify

$$p(x) = 4x$$

$$q(x) = 2$$

$$r(x) = 1$$

Using these three functions to calculate $Q(x)$ and $R(x)$ gives

$$Q(x) = (x - x_0)\frac{q(x)}{p(x)} = x\frac{2}{4x} = \frac{1}{2}$$

$$R(x) = (x - x_0)^2\frac{r(x)}{p(x)} = x^2\frac{1}{4x} = \frac{x}{4}$$

Because both denominators simplify to constants, we can see that both $Q(x)$ and $R(x)$ are analytic about $x_0 = 0$, so $x_0 = 0$ is a regular singular point.

Now we'll use

$$y = \sum_{n=0}^{\infty} c_n x^{n+r}$$

$$y' = \sum_{n=0}^{\infty} (n + r)c_n x^{n+r-1}$$

© Krista King Math

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1)c_n x^{n+r-2}$$

to make substitutions into the differential equation.

$$4xy'' + 2y' + y = 0$$

$$4x\sum_{n=0}^{\infty} (n+r)(n+r-1)c_n x^{n+r-2} + 2\sum_{n=0}^{\infty} (n+r)c_n x^{n+r-1} + \sum_{n=0}^{\infty} c_n x^{n+r} = 0$$

$$\sum_{n=0}^{\infty} 4(n+r)(n+r-1)c_n x^{n+r-1} + \sum_{n=0}^{\infty} 2(n+r)c_n x^{n+r-1} + \sum_{n=0}^{\infty} c_n x^{n+r} = 0$$

Combine terms with equivalent powers of x.

$$\sum_{n=0}^{\infty} 2(n+r)(2n+2r-1)c_n x^{n+r-1} + \sum_{n=0}^{\infty} c_n x^{n+r} = 0$$

$$x^r \left[\sum_{n=0}^{\infty} 2(n+r)(2n+2r-1)c_n x^{n-1} + \sum_{n=0}^{\infty} c_n x^n \right] = 0$$

The first series starts with the x^{-1} term, while the second series starts with the x^0 term, so we'll pull the x^{-1} term out of the first series.

$$x^r \left[2r(2r-1)c_0 x^{-1} + \sum_{n=1}^{\infty} 2(n+r)(2n+2r-1)c_n x^{n-1} + \sum_{n=0}^{\infty} c_n x^n \right] = 0$$

Now that the series are in phase, we'll substitute $k = n - 1$ and $n = k + 1$ into the first series, and $k = n$ into the second series.

© Krista King Math

$$x^r\left[2r(2r-1)c_0x^{-1}+\sum_{k=0}^{\infty}2(k+r+1)(2k+2r+1)c_{k+1}x^k+\sum_{k=0}^{\infty}c_kx^k\right]=0$$

Now that the series are in phase with matching indices, combine them.

$$x^r\left[2r(2r-1)c_0x^{-1}+\sum_{k=0}^{\infty}[2(k+r+1)(2k+2r+1)c_{k+1}+c_k]x^k\right]=0$$

This equation gives

$$2r(2r-1)c_0=0$$

$$2(k+r+1)(2k+2r+1)c_{k+1}+c_k=0 \qquad k=0,\ 1,\ 2,\ \ldots$$

or

$$r(2r-1)=0$$

$$c_{k+1}=-\frac{c_k}{2(k+r+1)(2k+2r+1)} \qquad k=0,\ 1,\ 2,\ \ldots$$

The indicial equation gives us $r_1=0$ and $r_2=1/2$. Substituting these indicial roots into the recurrence relation, we get

	For $r_1=1/2$	For $r_2=0$
	$c_{k+1}=-\dfrac{c_k}{2(k+\frac{3}{2})(2k+2)}$	$c_{k+1}=-\dfrac{c_k}{2(k+1)(2k+1)}$
$k=0$	$c_1=-\dfrac{c_0}{3!}$	$c_1=-\dfrac{c_0}{2!}$

© Krista King Math

$k = 1$	$c_2 = \frac{c_0}{5!}$	$c_2 = \frac{c_0}{4!}$
$k = 2$	$c_3 = -\frac{c_0}{7!}$	$c_3 = -\frac{c_0}{6!}$
...	...	...

Forming these coefficients into series around $x_0 = 0$ gives,

$$c_0(x - x_0)^0 + c_1(x - x_0)^1 + c_2(x - x_0)^2 + c_3(x - x_0)^3 + \dots$$

$$c_0(x - 0)^0 - \frac{1}{3!}c_0(x - 0)^1 + \frac{1}{5!}c_0(x - 0)^2 + \frac{1}{7!}c_0(x - 0)^3 - \dots$$

$$c_0\left(1 - \frac{1}{3!}x + \frac{1}{5!}x^2 - \frac{1}{7!}x^3 + \dots\right)$$

and

$$c_0(x - x_0)^0 + c_1(x - x_0)^1 + c_2(x - x_0)^2 + c_3(x - x_0)^3 + \dots$$

$$c_0(x - 0)^0 - \frac{1}{2!}c_0(x - 0)^1 + \frac{1}{4!}c_0(x - 0)^2 + \frac{1}{6!}c_0(x - 0)^3 + \dots$$

$$c_0\left(1 - \frac{1}{2!}x + \frac{1}{4!}x^2 - \frac{1}{6!}x^3 + \dots\right)$$

So the series solutions are

$$y_1(x) = c_0x^0\left(1 - \frac{1}{2!}x + \frac{1}{4!}x^2 - \frac{1}{6!}x^3 + \dots\right) = c_0\sum_{k=0}^{\infty}\frac{(-1)^k}{(2k)!}x^k$$

$$y_2(x) = c_0x^{\frac{1}{2}}\left(1 - \frac{1}{3!}x + \frac{1}{5!}x^2 - \frac{1}{7!}x^3 + \dots\right) = c_0x^{\frac{1}{2}}\sum_{k=0}^{\infty}\frac{(-1)^k}{(2k+1)!}x^k$$

© Krista King Math

and the general solution is

$$y(x) = C_1 y_1(x) + C_2 y_2(x)$$

$$y(x) = C_1 c_0 \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^k + C_2 c_0 x^{\frac{1}{2}} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^k$$

$$y(x) = C_1 \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^k + C_2 x^{\frac{1}{2}} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^k$$

■ 3. Determine the regularity of the singular point $x_0 = 0$ of the differential equation, use the method of Frobenius to build any solution(s) around that point, then find the general solution.

$$x^2y'' + x^2y' - 6y = 0$$

Solution:

Matching this differential equation to the standard form $p(x)y'' + q(x)y' + r(x)y = 0$, we can identify

$$p(x) = x^2$$

$$q(x) = x^2$$

$$r(x) = -6$$

Using these three functions to calculate $Q(x)$ and $R(x)$ gives

© Krista King Math

$$Q(x) = (x - x_0)\frac{q(x)}{p(x)} = x\frac{x^2}{x^2} = x$$

$$R(x) = (x - x_0)^2\frac{r(x)}{p(x)} = x^2\frac{-6}{x^2} = -6$$

Because both denominators simplify to constants, we can see that both $Q(x)$ and $R(x)$ are analytic about $x_0 = 0$, so $x_0 = 0$ is a regular singular point.

Now we'll use

$$y = \sum_{n=0}^{\infty} c_n x^{n+r}$$

$$y' = \sum_{n=0}^{\infty} (n + r)c_n x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} (n + r)(n + r - 1)c_n x^{n+r-2}$$

to make substitutions into the differential equation.

$$x^2y'' + x^2y' - 6y = 0$$

$$x^2\sum_{n=0}^{\infty} (n + r)(n + r - 1)c_n x^{n+r-2} + x^2\sum_{n=0}^{\infty} (n + r)c_n x^{n+r-1} - 6\sum_{n=0}^{\infty} c_n x^{n+r} = 0$$

$$\sum_{n=0}^{\infty} (n + r)(n + r - 1)c_n x^{n+r} + \sum_{n=0}^{\infty} (n + r)c_n x^{n+r+1} - \sum_{n=0}^{\infty} 6c_n x^{n+r} = 0$$

Combine terms with equivalent powers of x.

© Krista King Math

$$\sum_{n=0}^{\infty}[(n+r)(n+r-1)c_n-6c_n]x^{n+r}+\sum_{n=0}^{\infty}(n+r)c_nx^{n+r+1}=0$$

$$x^r\left[\sum_{n=0}^{\infty}[(n+r)(n+r-1)-6]c_nx^n+\sum_{n=0}^{\infty}(n+r)c_nx^{n+1}\right]=0$$

The first series starts with the x^0 term, while the second series starts with the x^1 term, so we'll pull the x^0 term out of the first series.

$$x^r\left[(r^2-r-6)c_0+\sum_{n=1}^{\infty}[(n+r)(n+r-1)-6]c_nx^n+\sum_{n=0}^{\infty}(n+r)c_nx^{n+1}\right]=0$$

Now that the series are in phase, we'll substitute $k=n+1$ and $n=k-1$ into the second series, and $k=n$ into the first series.

$$x^r\left[(r^2-r-6)c_0+\sum_{k=1}^{\infty}[(k+r)(k+r-1)-6]c_kx^k+\sum_{k=1}^{\infty}(k+r-1)c_{k-1}x^k\right]=0$$

Now that the series are in phase with matching indices, combine them.

$$x^r\left[(r^2-r-6)c_0+\sum_{k=1}^{\infty}[[(k+r)(k+r-1)-6]c_k+(k+r-1)c_{k-1}]x^k\right]=0$$

This equation gives

$$(r^2-r-6)c_0=0 \qquad k=0$$

$$[(k+r)(k+r-1)-6]c_k+(k+r-1)c_{k-1}=0 \qquad k=1,\ 2,\ 3,\ \ldots$$

or

© Krista King Math

$$(r-3)(r+2)=0 \qquad k=0$$

$$c_k=-\frac{(k+r-1)c_{k-1}}{(k+r)(k+r-1)-6} \qquad k=1,\,2,\,3,\,\ldots$$

The indicial equation gives us $r_1=3$ and $r_2=-2$. Substituting these indicial roots into the recurrence relation, we get

	For $r_1=3$	For $r_2=-2$
	$c_k=-\frac{(k+2)c_{k-1}}{(k+3)(k+2)-6}$	$c_k=-\frac{(k-3)c_{k-1}}{(k-2)(k-3)-6}$
$k=1$	$c_1=-\frac{c_0}{2}$	$c_1=-\frac{c_0}{2}$
$k=2$	$c_2=\frac{c_0}{7}$	$c_2=\frac{c_0}{12}$
$k=3$	$c_3=-\frac{5c_0}{168}$	$c_3=0$
...	...	...

Forming these coefficients into series around $x_0=0$ gives,

$$c_0(x-x_0)^0+c_1(x-x_0)^1+c_2(x-x_0)^2+c_3(x-x_0)^3+\ldots$$

$$c_0x^0-\frac{1}{2}c_0x^1+\frac{1}{7}c_0x^2-\frac{5}{168}c_0x^3+\ldots$$

$$c_0\left(1-\frac{1}{2}x+\frac{1}{7}x^2-\frac{5}{168}x^3+\ldots\right)$$

and

© Krista King Math

$$c_0(x-x_0)^0 + c_1(x-x_0)^1 + c_2(x-x_0)^2 + c_3(x-x_0)^3 + \dots$$

$$c_0 x^0 - \frac{1}{2}c_0 x^1 + \frac{1}{12}c_0 x^2 + \dots$$

$$c_0\left(1 - \frac{1}{2}x + \frac{1}{12}x^2 + \dots\right)$$

So the series solutions are

$$y_1(x) = c_0 x^3\left(1 - \frac{1}{2}x + \frac{1}{7}x^2 - \frac{5}{168}x^3 + \dots\right)$$

$$y_2(x) = c_0 x^{-2}\left(1 - \frac{1}{2}x + \frac{1}{12}x^2 + \dots\right)$$

and the general solution is

$$y(x) = C_1 y_1(x) + C_2 y_2(x)$$

$$y(x) = C_1 c_0 x^3\left(1 - \frac{1}{2}x + \frac{1}{7}x^2 - \dots\right)$$

$$+ C_2 c_0 x^{-2}\left(1 - \frac{1}{2}x + \frac{1}{12}x^2 + \dots\right)$$

$$y(x) = C_1 x^3\left(1 - \frac{1}{2}x + \frac{1}{7}x^2 - \dots\right) + C_2 x^{-2}\left(1 - \frac{1}{2}x + \frac{1}{12}x^2 + \dots\right)$$

© Krista King Math

■ 4. Determine the regularity of the singular point $x_0 = 0$ of the differential equation, use the method of Frobenius to build any solution(s) around that point, then find the general solution.

$$8x^2y'' + x^2y' + (2 - x)y = 0$$

Solution:

Matching this differential equation to the standard form $p(x)y'' + q(x)y' + r(x)y = 0$, we can identify

$$p(x) = 8x^2$$

$$q(x) = x^2$$

$$r(x) = 2 - x$$

Using these three functions to calculate $Q(x)$ and $R(x)$ gives

$$Q(x) = (x - x_0)\frac{q(x)}{p(x)} = x\frac{x^2}{8x^2} = \frac{x}{8}$$

$$R(x) = (x - x_0)^2\frac{r(x)}{p(x)} = x^2\frac{2 - x}{8x^2} = \frac{2 - x}{8}$$

Because both denominators simplify to constants, we can see that both $Q(x)$ and $R(x)$ are analytic about $x_0 = 0$, so $x_0 = 0$ is a regular singular point.

Now we'll use

© Krista King Math



$$y = \sum_{n=0}^{\infty} c_n x^{n+r}$$

$$y' = \sum_{n=0}^{\infty} (n+r)c_n x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1)c_n x^{n+r-2}$$

to make substitutions into the differential equation.

$$8x^2y'' + x^2y' + (2-x)y = 0$$

$$8x^2 \sum_{n=0}^{\infty} (n+r)(n+r-1)c_n x^{n+r-2} + x^2 \sum_{n=0}^{\infty} (n+r)c_n x^{n+r-1}$$

$$+(2-x) \sum_{n=0}^{\infty} c_n x^{n+r} = 0$$

$$\sum_{n=0}^{\infty} 8(n+r)(n+r-1)c_n x^{n+r} + \sum_{n=0}^{\infty} (n+r)c_n x^{n+r+1}$$

$$+ \sum_{n=0}^{\infty} 2c_n x^{n+r} - \sum_{n=0}^{\infty} c_n x^{n+r+1} = 0$$

Combine terms with equivalent powers of x.

$$\sum_{n=0}^{\infty} (8(n+r)(n+r-1)+2)c_n x^{n+r} + \sum_{n=0}^{\infty} (n+r-1)c_n x^{n+r+1} = 0$$

© Krista King Math

$$x^r\left[\sum_{n=0}^{\infty}(8(n+r)(n+r-1)+2)c_n x^n+\sum_{n=0}^{\infty}(n+r-1)c_n x^{n+1}\right]=0$$

The first series starts with the x^0 term, while the second series starts with the x^1 term, so we'll pull the x^0 term out of the first series.

$$x^r\left[(8r(r-1)+2)c_0+\sum_{n=1}^{\infty}(8(n+r)(n+r-1)+2)c_n x^n+\sum_{n=0}^{\infty}(n+r-1)c_n x^{n+1}\right]=0$$

Now that the series are in phase, we'll substitute $k=n+1$ and $n=k-1$ into the second series, and $k=n$ into the first series.

$$x^r\left[2(4r^2-4r+1)c_0+\sum_{k=1}^{\infty}(8(k+r)(k+r-1)+2)c_k x^k+\sum_{k=1}^{\infty}(k+r-2)c_{k-1}x^k\right]=0$$

Now that the series are in phase with matching indices, combine them.

$$x^r\left[2(2r-1)^2c_0+\sum_{k=1}^{\infty}[(8(k+r)(k+r-1)+2)c_k+(k+r-2)c_{k-1}]x^k\right]=0$$

This equation gives

$$2(2r-1)^2c_0=0 \qquad k=0$$

$$(8(k+r)(k+r-1)+2)c_k+(k+r-2)c_{k-1}=0 \qquad k=1,\,2,\,3,\,\ldots$$

or

$$(2r-1)^2=0 \qquad k=0$$

$$c_k=-\frac{(k+r-2)c_{k-1}}{8(k+r)(k+r-1)+2} \qquad k=1,\,2,\,3,\,\ldots$$

© Krista King Math

The indicial equation gives us $r_1 = r_2 = 1/2$. Substituting this indicial root into the recurrence relation, we get

For $r_1 = r_2 = 1/2$

$$c_k = -\frac{(k-\frac{3}{2})c_{k-1}}{8(k+\frac{1}{2})(k-\frac{1}{2})+2}$$

$$c_k = -\frac{(2k-3)c_{k-1}}{16k^2}$$

$k = 1$	$c_1 = \frac{1}{16}c_0$
$k = 2$	$c_2 = -\frac{1}{1{,}024}c_0$
$k = 3$	$c_3 = \frac{1}{49{,}152}c_0$
...	...

Forming these coefficients into series around $x_0 = 0$ gives,

$$c_0(x-x_0)^0 + c_1(x-x_0)^1 + c_2(x-x_0)^2 + c_3(x-x_0)^3 + \dots$$

$$c_0x^0 + \frac{1}{16}c_0x^1 - \frac{1}{1{,}024}c_0x^2 + \frac{1}{49{,}152}c_0x^3 + \dots$$

$$c_0\left(1 + \frac{1}{16}x - \frac{1}{1{,}024}x^2 + \frac{1}{49{,}152}x^3 + \dots\right)$$

So the series solutions are

© Krista King Math

$$y_1(x) = c_0 x^{\frac{1}{2}} \left(1 + \frac{1}{16}x - \frac{1}{1{,}024}x^2 + \frac{1}{49{,}152}x^3 + \dots\right)$$

and

$$y_2(x) = y_1(x)\ln x + \sum_{n=0}^{\infty} b_n x^{n+r_1}$$

$$y_2(x) = c_0 x^{\frac{1}{2}} \ln x \left(1 + \frac{1}{16}x - \frac{1}{1{,}024}x^2 + \frac{1}{49{,}152}x^3 + \dots\right) + \sum_{n=0}^{\infty} b_n x^{n+\frac{1}{2}}$$

and the general solution is

$$y(x) = C_1 y_1(x) + C_2 y_2(x)$$

$$y(x) = C_1 c_0 x^{\frac{1}{2}} \left(1 + \frac{1}{16}x - \frac{1}{1{,}024}x^2 + \frac{1}{49{,}152}x^3 + \dots\right)$$

$$+ C_2 \left(c_0 x^{\frac{1}{2}} \ln x \left(1 + \frac{1}{16}x - \frac{1}{1{,}024}x^2 + \frac{1}{49{,}152}x^3 + \dots\right) + \sum_{n=0}^{\infty} b_n x^{n+\frac{1}{2}}\right)$$

$$y(x) = C_1 x^{\frac{1}{2}} \left(1 + \frac{1}{16}x - \frac{1}{1{,}024}x^2 + \frac{1}{49{,}152}x^3 + \dots\right)$$

$$+ C_2 x^{\frac{1}{2}} \ln x \left(1 + \frac{1}{16}x - \frac{1}{1{,}024}x^2 + \frac{1}{49{,}152}x^3 + \dots\right) + C_2 \sum_{n=0}^{\infty} b_n x^{n+\frac{1}{2}}$$

■ 5. Determine the regularity of the singular point $x_0 = 0$ of the differential equation, use the method of Frobenius to build any solution(s) around that point, then find the general solution.

© Krista King Math



$$3x^2y'' + 3xy' - 2xy = 0$$

Solution:

Matching this differential equation to the standard form $p(x)y'' + q(x)y' + r(x)y = 0$, we can identify

$$p(x) = 3x^2$$

$$q(x) = 3x$$

$$r(x) = -2x$$

Using these three functions to calculate $Q(x)$ and $R(x)$ gives

$$Q(x) = (x - x_0)\frac{q(x)}{p(x)} = x\frac{3x}{3x^2} = 1$$

$$R(x) = (x - x_0)^2\frac{r(x)}{p(x)} = x^2\frac{-2x}{3x^2} = -\frac{2x}{3}$$

Because both denominators simplify to constants, we can see that both $Q(x)$ and $R(x)$ are analytic about $x_0 = 0$, so $x_0 = 0$ is a regular singular point.

Now we'll use

$$y = \sum_{n=0}^{\infty} c_n x^{n+r}$$

$$y' = \sum_{n=0}^{\infty} (n + r)c_n x^{n+r-1}$$

© Krista King Math

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1)c_n x^{n+r-2}$$

to make substitutions into the differential equation.

$$3x^2y'' + 3xy' - 2xy = 0$$

$$3x^2 \sum_{n=0}^{\infty} (n+r)(n+r-1)c_n x^{n+r-2} + 3x \sum_{n=0}^{\infty} (n+r)c_n x^{n+r-1}$$

$$-2x \sum_{n=0}^{\infty} c_n x^{n+r} = 0$$

$$\sum_{n=0}^{\infty} 3(n+r)(n+r-1)c_n x^{n+r} + \sum_{n=0}^{\infty} 3(n+r)c_n x^{n+r} - \sum_{n=0}^{\infty} 2c_n x^{n+r+1} = 0$$

Combine terms with equivalent powers of x.

$$\sum_{n=0}^{\infty} (3(n+r)(n+r-1) + 3(n+r))c_n x^{n+r} - \sum_{n=0}^{\infty} 2c_n x^{n+r+1} = 0$$

$$\sum_{n=0}^{\infty} 3(n+r)((n+r-1)+1)c_n x^{n+r} - \sum_{n=0}^{\infty} 2c_n x^{n+r+1} = 0$$

$$\sum_{n=0}^{\infty} 3(n+r)^2 c_n x^{n+r} - \sum_{n=0}^{\infty} 2c_n x^{n+r+1} = 0$$

$$x^r \left[\sum_{n=0}^{\infty} 3(n+r)^2 c_n x^n - \sum_{n=0}^{\infty} 2c_n x^{n+1} \right] = 0$$

© Krista King Math

The first series starts with the x^0 term, while the second series starts with the x^1 term, so we'll pull the x^0 term out of the first series.

$$x^r \left[3r^2 c_0 + \sum_{n=1}^{\infty} 3(n+r)^2 c_n x^n - \sum_{n=0}^{\infty} 2c_n x^{n+1} \right] = 0$$

Now that the series are in phase, we'll substitute $k = n + 1$ and $n = k - 1$ into the second series, and $k = n$ into the first series.

$$x^r \left[3r^2 c_0 + \sum_{k=1}^{\infty} 3(k+r)^2 c_k x^k - \sum_{k=1}^{\infty} 2c_{k-1} x^k \right] = 0$$

Now that the series are in phase with matching indices, combine them.

$$x^r \left[3r^2 c_0 + \sum_{k=1}^{\infty} [3(k+r)^2 c_k - 2c_{k-1}] x^k \right] = 0$$

This equation gives

$$3r^2 c_0 = 0 \qquad k = 0$$

$$3(k+r)^2 c_k - 2c_{k-1} = 0 \qquad k = 1, 2, 3, \ldots$$

or

$$r_1 = r_2 = 0 \qquad k = 0$$

$$c_k = \frac{2c_{k-1}}{3(k+r)^2} \qquad k = 1, 2, 3, \ldots$$

The indicial equation gives us $r_1 = r_2 = 0$. Substituting this indicial root into the recurrence relation, we get

© Krista King Math



For $r_1 = r_2 = 0$

$$c_{k+1} = \frac{2c_k}{3(k+1)^2}$$

$k = 1$ $\quad c_2 = \frac{2^2}{3^2 \cdot 2^2} c_0$

$k = 2$ $\quad c_3 = \frac{2^3}{3^3 \cdot 3^2 \cdot 2^2} c_0$

... $\quad$ ...

Forming these coefficients into series around $x_0 = 0$ gives,

$$c_0(x-0)^0 + \frac{2}{3}c_0(x-0)^1 + \frac{2^2}{3^2(2!)^2}c_0(x-0)^2 + \ldots$$

So the series solutions are

$$y_1(x) = \sum_{k=0}^{\infty} \frac{2^k}{3^k((k+1)!)^2} c_0 \cdot x^k = c_0 \sum_{k=0}^{\infty} \frac{2^k}{3^k((k+1)!)^2} x^k$$

$$y_2(x) = c_0 \ln x \sum_{k=0}^{\infty} \frac{2^k}{3^k((k+1)!)^2} x^k + b_0 \sum_{k=0}^{\infty} \frac{2^k}{3^k((k+1)!)^2} x^k$$

and the general solution is

$$y(x) = C_1 y_1(x) + C_2 y_2(x)$$

$$y(x) = C_1 c_0 \sum_{k=0}^{\infty} \frac{2^k}{3^k((k+1)!)^2} x^k + C_2 c_0 \ln x \sum_{k=0}^{\infty} \frac{2^k}{3^k((k+1)!)^2} x^k$$

© Krista King Math

$$+C_2 b_0 \sum_{k=0}^{\infty} \frac{2^k}{3^k((k+1)!)^2} x^k$$

$$y(x) = C_1 \sum_{k=0}^{\infty} \frac{2^k}{3^k((k+1)!)^2} x^k + C_2 \ln x \sum_{k=0}^{\infty} \frac{2^k}{3^k((k+1)!)^2} x^k$$

$$+C_3 \sum_{k=0}^{\infty} \frac{2^k}{3^k((k+1)!)^2} x^k$$

■ 6. Determine the regularity of the singular point $x_0 = 0$ of the differential equation, use the method of Frobenius to build any solution(s) around that point, then find the general solution.

$$3xy'' + 2y' + 2y = 0$$

Solution:

Matching this differential equation to the standard form $p(x)y'' + q(x)y' + r(x)y = 0$, we can identify

$$p(x) = 3x$$

$$q(x) = 2$$

$$r(x) = 2$$

Using these three functions to calculate $Q(x)$ and $R(x)$ gives

© Krista King Math

$$Q(x) = (x - x_0)\frac{q(x)}{p(x)} = x\frac{2}{3x} = \frac{2}{3}$$

$$R(x) = (x - x_0)^2\frac{r(x)}{p(x)} = x^2\frac{2}{3x} = \frac{2x}{3}$$

Because the denominators of $Q(x)$ and $R(x)$ are constants ($(2/3)/1$ and $(2x/3)/1$), these functions are analytic about $x_0 = 0$, so $x_0 = 0$ is a regular singular point.

Now we'll use

$$y = \sum_{n=0}^{\infty} c_n x^{n+r}$$

$$y' = \sum_{n=0}^{\infty} (n + r)c_n x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} (n + r)(n + r - 1)c_n x^{n+r-2}$$

to make substitutions into the differential equation.

$$3xy'' + 2y' + 2y = 0$$

$$3x\sum_{n=0}^{\infty} (n + r)(n + r - 1)c_n x^{n+r-2} + 2\sum_{n=0}^{\infty} (n + r)c_n x^{n+r-1} + 2\sum_{n=0}^{\infty} c_n x^{n+r} = 0$$

$$\sum_{n=0}^{\infty} 3(n + r)(n + r - 1)c_n x^{n+r-1} + \sum_{n=0}^{\infty} 2(n + r)c_n x^{n+r-1} + \sum_{n=0}^{\infty} 2c_n x^{n+r} = 0$$

Combine terms with equivalent powers of x.

© Krista King Math

$$\sum_{n=0}^{\infty}[3(n+r)(n+r-1)+2(n+r)]c_n x^{n+r-1}+\sum_{n=0}^{\infty}2c_n x^{n+r}=0$$

$$\sum_{n=0}^{\infty}(n+r)(3n+3r-1)c_n x^{n+r-1}+\sum_{n=0}^{\infty}2c_n x^{n+r}=0$$

$$x^r\left[\sum_{n=0}^{\infty}(n+r)(3n+3r-1)c_n x^{n-1}+\sum_{n=0}^{\infty}2c_n x^{n}\right]=0$$

The first series starts with the x^{-1} term and the second series starts with the x^0 term, so we'll pull the x^{-1} term out of the first series,

$$x^r\left[r(3r-1)c_0x^{-1}+\sum_{n=1}^{\infty}(n+r)(3n+3r-1)c_n x^{n-1}+\sum_{n=0}^{\infty}2c_n x^{n}\right]=0$$

Now that the series are in phase, we'll substitute $k=n+1$ and $n=k-1$ into the first series and $k=n$ into the first series.

$$x^r\left[r(3r-1)c_0x^{-1}+\sum_{k=1}^{\infty}(k+r)(3k+3r-1)c_k x^{k-1}+\sum_{k=1}^{\infty}2c_{k-1} x^{k-1}\right]=0$$

Now that the series are in phase with matching indices, combine them.

$$x^r\left[r(3r-1)c_0x^{-1}+\sum_{k=1}^{\infty}[(k+r)(3k+3r-1)c_k+2c_{k-1}]x^{k-1}\right]=0$$

This equation gives

$$r(3r-1)c_0=0 \qquad k=0$$

$$(k+r)(3k+3r-1)c_k+2c_{k-1}=0 \qquad k=1,\,2,\,3,\,\ldots$$

© Krista King Math

or

$$r(3r-1)=0 \qquad k=0$$

$$c_k=-\frac{2c_{k-1}}{(k+r)(3k+3r-1)} \qquad k=1,\ 2,\ 3,\ \ldots$$

The indicial equation gives us $r_1=0$ and $r_2=1/3$. Substituting these indicial roots into the recurrence relation, we get

	For $r=0$	For $r=1/3$
	$c_k=-\frac{2c_{k-1}}{(k+r)(3k+3r-1)}$	$c_k=-\frac{2c_{k-1}}{(k+r)(3k+3r-1)}$
$k=1$	$c_1=-c_0$	$c_1=-\frac{c_0}{2}$
$k=2$	$c_2=\frac{c_0}{5}$	$c_2=\frac{c_0}{14}$
$k=3$	$c_3=-\frac{c_0}{60}$	$c_3=-\frac{c_0}{210}$
$k=4$	$c_4=\frac{c_0}{1{,}320}$	$c_4=\frac{c_0}{5{,}460}$
...	...	

Forming these coefficients into series around $x_0=0$ gives,

$$c_0(x-x_0)^0+c_1(x-x_0)^1+c_2(x-x_0)^2+c_3(x-x_0)^3+\ldots$$

$$c_0-c_0x+\frac{c_0}{5}x^2-\frac{c_0}{60}x^3+\frac{c_0}{1{,}320}x^4-\ldots$$

© Krista King Math

$$c_0\left(1-x+\frac{1}{5}x^2-\frac{1}{60}x^3+\frac{1}{1{,}320}x^4-\ldots\right)$$

and

$$c_0(x-x_0)^0+c_1(x-x_0)^1+c_2(x-x_0)^2+c_3(x-x_0)^3+\ldots$$

$$c_0-\frac{c_0}{2}x+\frac{c_0}{14}x^2-\frac{c_0}{210}x^3+\frac{c_0}{5{,}460}x^4+\ldots$$

$$c_0\left(1-\frac{1}{2}x+\frac{1}{14}x^2-\frac{1}{210}x^3+\frac{1}{5{,}460}x^4+\ldots\right)$$

So the series solutions are

$$y_1(x)=c_0\left(1-x+\frac{1}{5}x^2-\frac{1}{60}x^3+\frac{1}{1{,}320}x^4-\ldots\right)$$

$$y_2(x)=c_0\left(1-\frac{1}{2}x+\frac{1}{14}x^2-\frac{1}{210}x^3+\frac{1}{5{,}460}x^4+\ldots\right)$$

and the general solution is

$$y(x)=C_1y_1(x)+C_2y_2(x)$$

$$y(x)=C_1c_0\left(1-x+\frac{1}{5}x^2-\frac{1}{60}x^3+\frac{1}{1{,}320}x^4-\ldots\right)$$

$$+C_2c_0\left(1-\frac{1}{2}x+\frac{1}{14}x^2-\frac{1}{210}x^3+\frac{1}{5{,}460}x^4+\ldots\right)$$

$$y(x)=C_1\left(1-x+\frac{1}{5}x^2-\frac{1}{60}x^3+\frac{1}{1{,}320}x^4-\ldots\right)$$

© Krista King Math

$$+C_2\left(1-\frac{1}{2}x+\frac{1}{14}x^2-\frac{1}{210}x^3+\frac{1}{5{,}460}x^4+\ldots\right)$$

© Krista King Math

FOR THE LOVE OF MATH

WWW.KRISTAKINGMATH.COM